

## **CC4003NI Fundamentals of Robotics and IOT**

**50% Group Coursework**

**Submission: Final Submission**

**Academic Semester: AY 2024/25**

**Credit: 30 credit year long module**

| Group Name: |                            |                |               |
|-------------|----------------------------|----------------|---------------|
| SN          | Student Name               | College ID     | University ID |
| 1           | Erick Pradhan (AI4) Leader | NP01AI4A240097 | 24046582      |
| 2           | Papit Ghimire (AI4)        | NP01AI4A240088 | 24046608      |
| 3           | Saket Yadav (AI4)          | NP01AI4A240109 | 24046630      |
| 4           | Sangam Pandy (AI4)         | NP01AI4A240113 | 24046637      |

**Assignment Due Date: Monday, May 12, 2025**

**Assignment Submission Date: Monday, May 12, 2025**

**Word Count: 3049**

**Submitted to: Mr. Shishir Subedi**

*I confirm that I understand my coursework needs to be submitted online via MST Classroom under the relevant module page before the deadline in order for my assignment to be accepted and marked. I am fully aware that late submissions will be treated as non-submission and a mark of zero will be awarded.*

# Turnitin Report



Page 1 of 29 - Cover Page

Submission ID trn:oid::3618:95371547

## 24046582 ERICK PRADHAN.docx

Islington College, Nepal

### Document Details

Submission ID

trn:oid::3618:95368708

Submission Date

May 11, 2025, 11:30 PM GMT+5:45

Download Date

May 12, 2025, 12:24 AM GMT+5:45

File Name

24046582 ERICK PRADHAN.docx

File Size

26.3 KB

30 Pages

4,123 Words

22,590 Characters



Page 2 of 35 - Integrity Overview

Submission ID trn:oid::3618:95368708

## 14% Overall Similarity

The combined total of all matches, including overlapping sources, for each database.

### Match Groups

-  **55 Not Cited or Quoted 11%**  
Matches with neither in-text citation nor quotation marks
-  **7 Missing Quotations 3%**  
Matches that are still very similar to source material
-  **0 Missing Citation 0%**  
Matches that have quotation marks, but no in-text citation
-  **0 Cited and Quoted 0%**  
Matches with in-text citation present, but no quotation marks

### Top Sources

- 4%  Internet sources
- 1%  Publications
- 11%  Submitted works (Student Papers)

### Integrity Flags

0 Integrity Flags for Review

Our system's algorithms look deeply at a document for any inconsistencies that would set it apart from a normal submission. If we notice something strange, we flag it for you to review.

A Flag is not necessarily an indicator of a problem. However, we'd recommend you focus your attention there for further review.

## Match Groups

- **55 Not Cited or Quoted 11%**  
Matches with neither in-text citation nor quotation marks
- **7 Missing Quotations 3%**  
Matches that are still very similar to source material
- **0 Missing Citation 0%**  
Matches that have quotation marks, but no in-text citation
- **0 Cited and Quoted 0%**  
Matches with in-text citation present, but no quotation marks

## Top Sources

- 4% Internet sources
- 1% Publications
- 11% Submitted works (Student Papers)

## Top Sources

The sources with the highest number of matches within the submission. Overlapping sources will not be displayed.

|    |                 |                                                                                   |     |
|----|-----------------|-----------------------------------------------------------------------------------|-----|
| 1  | Submitted works | Manchester Metropolitan University on 2023-05-12                                  | <1% |
| 2  | Submitted works | University of Greenwich on 2015-06-10                                             | <1% |
| 3  | Internet        | www.ncf.org.np                                                                    | <1% |
| 4  | Internet        | www2.mdpi.com                                                                     | <1% |
| 5  | Internet        | www.preprints.org                                                                 | <1% |
| 6  | Submitted works | University College Technology Sarawak on 2020-07-31                               | <1% |
| 7  | Submitted works | Asia Pacific University College of Technology and Innovation (UCTI) on 2025-03-20 | <1% |
| 8  | Submitted works | University of Wales Institute, Cardiff on 2019-09-04                              | <1% |
| 9  | Submitted works | Middle East College of Information Technology on 2020-01-27                       | <1% |
| 10 | Submitted works | Universiti Teknologi Malaysia on 2024-07-01                                       | <1% |

|    |                 |                                                                                    |     |
|----|-----------------|------------------------------------------------------------------------------------|-----|
| 11 | Submitted works | Coventry University on 2025-02-21                                                  | <1% |
| 12 | Submitted works | University of Bolton on 2021-07-23                                                 | <1% |
| 13 | Submitted works | Middlesex University on 2023-04-05                                                 | <1% |
| 14 | Submitted works | University of Liverpool on 2025-03-16                                              | <1% |
| 15 | Submitted works | University of Nottingham on 2025-02-27                                             | <1% |
| 16 | Submitted works | Kingston University on 2025-04-10                                                  | <1% |
| 17 | Submitted works | Nottingham Trent University on 2023-04-28                                          | <1% |
| 18 | Submitted works | University of Ulster on 2024-08-23                                                 | <1% |
| 19 | Submitted works | Westcliff University on 2025-03-19                                                 | <1% |
| 20 | Submitted works | Napier University on 2024-12-04                                                    | <1% |
| 21 | Internet        | pdfslide.net                                                                       | <1% |
| 22 | Submitted works | Kingston University on 2025-03-29                                                  | <1% |
| 23 | Submitted works | University of Westminster on 2017-04-17                                            | <1% |
| 24 | Submitted works | Institute of Research & Postgraduate Studies, Universiti Kuala Lumpur on 2025-0... | <1% |

|    |                 |                                                                                      |     |
|----|-----------------|--------------------------------------------------------------------------------------|-----|
| 25 | Internet        | repository.psa.edu.my                                                                | <1% |
| 26 | Submitted works | Brunel University on 2025-01-30                                                      | <1% |
| 27 | Submitted works | Informatics Education Limited on 2012-03-28                                          | <1% |
| 28 | Internet        | ujae.org.ua                                                                          | <1% |
| 29 | Submitted works | Athlone Institute of Technology on 2020-04-27                                        | <1% |
| 30 | Submitted works | Polytechnics Mauritius on 2024-04-22                                                 | <1% |
| 31 | Publication     | T. Vasudeva Reddy, K. Madhava Rao. "Recent Trends in VLSI and Semiconductor P...     | <1% |
| 32 | Submitted works | Universiti Teknologi MARA on 2018-12-27                                              | <1% |
| 33 | Submitted works | University of Greenwich on 2024-04-10                                                | <1% |
| 34 | Submitted works | University of Ulster on 2025-05-09                                                   | <1% |
| 35 | Submitted works | Jaypee University of Information Technology on 2018-05-11                            | <1% |
| 36 | Publication     | V. Sharmila, S. Kannadhasan, A. Rajiv Kannan, P. Sivakumar, V. Vennila. "Challeng... | <1% |
| 37 | Submitted works | Queensland Academy for Science, Maths and Technology on 2023-11-12                   | <1% |
| 38 | Submitted works | University of Surrey on 2025-05-09                                                   | <1% |

## **Acknowledgement**

We would like to give our heartfelt gratitude to Islington College for giving us this opportunity to learn explore new things in this very subject. As we are very indebt towards there good care to us.

Finally, we cannot express our gratitude towards our module leader and tutor Mr. Shishir Subedi for his way of teaching and giving us time to develop our skills for the wellbeing of the upcoming future. Your fun way of teaching and strict way of speaking motivated us throughout this project. We cannot describe our joy of being your subordinate students we are in debt and very grateful to have a tutor like you.

## **Abstract**

The project is about IOT Based Smart Agricultural System, specifically focusing on the development of an intelligent farming system that optimizes irrigation and environmental monitoring.

The objective is to work on designing and implementing an automated system that ensures efficient result, management and improved crop health by using IoT components such as sensors and cloud based platforms. The project aims to provide real time monitoring and data driven decision making for farmers.

## Table of contents

|        |                              |    |
|--------|------------------------------|----|
| 1.     | Introduction .....           | 1  |
| 1.1    | Current Scenario .....       | 3  |
| 1.2    | Problem Statement .....      | 4  |
| 1.3    | Project as a solution.....   | 5  |
| 1.4    | Aim and Objectives .....     | 6  |
| 2      | Background .....             | 7  |
| 2.1    | System Overview .....        | 7  |
| 2.2    | Design Diagrams .....        | 8  |
| 2.2.1. | Hardware Architecture: ..... | 8  |
| 2.2.2. | Circuit Diagram: .....       | 9  |
| 2.2.3. | Flowchart: .....             | 9  |
| 2.3    | Requirement Analysis .....   | 11 |
| 2.3.1  | Hardware Components .....    | 11 |
| 2.3.2  | Software Components .....    | 15 |
| 3      | Case Study .....             | 17 |
| 3.1    | Introduction .....           | 17 |
| 3.2    | Problem Statement .....      | 18 |
| 3.3    | System Analysis.....         | 19 |
| 3.4    | Recommendations .....        | 20 |
| 4      | Development .....            | 22 |
| 4.1    | Planning and Designs .....   | 22 |
| 4.2    | System Development .....     | 23 |
| 4.2.1  | Phase 1.....                 | 23 |
| 4.2.2  | Phase 2.....                 | 24 |



|       |                                                     |    |
|-------|-----------------------------------------------------|----|
| 4.2.3 | Phase 3.....                                        | 25 |
| 4.2.4 | Phase 4.....                                        | 26 |
| 4.2.5 | Phase 5.....                                        | 27 |
| 4.2.6 | Phase 6.....                                        | 29 |
| 4.2.7 | Pin Connections among the Hardware Components ..... | 30 |
| 5     | Results and Findings.....                           | 32 |
| 5.1   | Result.....                                         | 32 |
| 5.2   | Testing .....                                       | 33 |
| 5.2.1 | Test 1:.....                                        | 33 |
| 5.2.2 | Test 2:.....                                        | 35 |
| 5.2.3 | Test 3:.....                                        | 37 |
| 5.2.4 | Test 4:.....                                        | 40 |
| 5.2.5 | Test 5:.....                                        | 42 |
| 5.2.6 | Test 6:.....                                        | 44 |
| 6     | Future works .....                                  | 46 |
| 7     | Conclusion .....                                    | 47 |
| 8     | References.....                                     | 49 |
| 9     | Appendix .....                                      | 51 |
| 9.1   | Appendix A: Source Code.....                        | 51 |
| 9.2.  | Appendix B: Screenshots of the System .....         | 55 |
| 9.3.  | Appendix C: Contribution .....                      | 56 |

## Table of Figures

|                                                                                                    |    |
|----------------------------------------------------------------------------------------------------|----|
| Figure 1 – Advance IoT System.....                                                                 | 1  |
| Figure 2 – Agriculture's Contribution to GDP .....                                                 | 3  |
| Figure 3 – Problems .....                                                                          | 4  |
| Figure 4 – Hardware Architecture .....                                                             | 8  |
| Figure 5 – Circuit Diagram .....                                                                   | 9  |
| Figure 6 – Flowchart .....                                                                         | 10 |
| Figure 7 – ESP-32 Microcontroller .....                                                            | 11 |
| Figure 8 – Soil Moisture Sensor.....                                                               | 11 |
| Figure 9 – DHT11 Temperature and Humidity Sensor .....                                             | 11 |
| Figure 10 – Ultrasonic Sensor (HC-SR04).....                                                       | 12 |
| Figure 11 – Water Pump .....                                                                       | 12 |
| Figure 12 – Buzzer .....                                                                           | 12 |
| Figure 13 – 5V Relay Module .....                                                                  | 13 |
| Figure 14 – I2C LCD .....                                                                          | 13 |
| Figure 15 – Jumper Wires.....                                                                      | 13 |
| Figure 16 – Breadboard .....                                                                       | 14 |
| Figure 17 – 5V 2A DC Adapter .....                                                                 | 14 |
| Figure 18 – Arduino IDE.....                                                                       | 15 |
| Figure 19 – Blynk .....                                                                            | 15 |
| Figure 20 – Fritzing .....                                                                         | 16 |
| Figure 21 – Cases of Precision Farming using IoT.....                                              | 17 |
| Figure 22 – Major Problems in Agriculture.....                                                     | 18 |
| Figure 23 – System Analysis.....                                                                   | 19 |
| Figure 24 – Advance IoT.....                                                                       | 20 |
| Figure 25 – Phase 1_Connecting DHT11, Breadboard and ESP-32 .....                                  | 23 |
| Figure 26 – Phase 2_Connecting Ultrasonic sensor with the rest of the circuits.....                | 24 |
| Figure 27 – Phase 3_Connecting buzzer and soil moisture sensor with the rest of the circuits ..... | 25 |

|                                                                                                                     |    |
|---------------------------------------------------------------------------------------------------------------------|----|
| Figure 28 – Phase 4_Connecting I2C LCD with the rest of the circuits .....                                          | 26 |
| Figure 29 – Phase 5_Connecting 3V relay module, Water pump and 5V 2A DC Adapter with the rest of the circuits ..... | 28 |
| Figure 30 – Phase 6_Conneting Blynk .....                                                                           | 29 |
| Figure 31 – Test 1_Connecting DHT sensor with ESP-32 .....                                                          | 34 |
| Figure 32 – Test 1_Testing DHT sensor with ESP-32 .....                                                             | 34 |
| Figure 33 – Test 2_Conneting Ultrasonic sensor with the rest of the circuits .....                                  | 35 |
| Figure 34 – Test 2_Testing Ultrasonic sensor with the rest of the circuits .....                                    | 36 |
| Figure 35 – Test 3_Connecting Buzzer and Soil moisture sensor with the rest of the circuits .....                   | 38 |
| Figure 36 – Test 3_Testing Buzzer and Soil moisture sensor with the rest of the circuits .....                      | 39 |
| Figure 37 – Test 4_Connecting I2C LCD with the rest of the circuits .....                                           | 41 |
| Figure 38 – Test 4_Testing I2C LCD with the rest of the circuits .....                                              | 41 |
| Figure 39 – Test 5_Connecting 3V Relay module, Water pump and 5V DC Adapter with the rest of the circuits .....     | 42 |
| Figure 40 – Test 5_Testing 3V Relay module, Water pump and 5V DC Adapter with the rest of the circuits .....        | 43 |
| Figure 41 - Test 6_Testing the whole circuit with blynk .....                                                       | 44 |
| Figure 42 – Test 6_Connecting with Blynk application .....                                                          | 45 |
| Figure 43 – Benefits of using IoT in Agriculture .....                                                              | 47 |
| Figure 44 – Block Diagram of the System .....                                                                       | 55 |

## List of Tables

|                                         |    |
|-----------------------------------------|----|
| Table 1 – Pin Connections .....         | 30 |
| Table 2 – Test 1 .....                  | 33 |
| Table 3 – Test 2 .....                  | 35 |
| Table 4 – Test 3 .....                  | 37 |
| Table 5 – Test 4 .....                  | 40 |
| Table 6 – Test 5 .....                  | 42 |
| Table 7 – Test 6 .....                  | 44 |
| Table 8 – Individual Contribution ..... | 56 |

## 1. Introduction

Internet of Things (IoT) is a developing concept that connects everyday objects and devices to the internet enables automated data collection. This technology has advanced various industries and agriculture as well. By integrating IoT into farming we can create a great intelligent system that enhances productivity which also conserves resources and make real-time decisions based on environmental conditions. Whereas, sensors, cloud computing, and automated controls are some of the core components of the IoT-Based agricultural System that help optimize irrigation, monitor soil and manage climate effectively (Zhang et al., 2021).



Figure 1 – Advance IoT System

In agriculture, IoT enables farmers to do remote monitoring to their fields using smart sensors, actuators and devices. These sensors collect data of sensor (soil moisture, temperature and humidity) which then helps the farmer in automating irrigation and improving water efficiency (To minimize the waste of water). The implementation of IoT-based agricultural systems devices will not only reduce human labour but will also minimize crop loss due to the unpredictable weather condition. As the global demand for food increases in the whole world the adoption of IoT in agriculture becomes very crucial which will then help in ensuring sustainable and efficient farming practices.

Therefore, IoT Based Agricultural System farming solutions combine AI and ML to analyze the historical agricultural data and predict the future of Nepal by leveraging AI-driven analytics farmers can make precise decisions about irrigation schedules, pest control, and crop rotations. This predictive capability allows for better risk management which also utilize resources ensuring that agricultural practices remain both profitable and environmentally sustainable (Anand, 2023).

## 1.1 Current Scenario

Agriculture plays a very significant role not only in Nepal's economy but also in many other countries, employing approximately 66% of the population and contributing nearly 27% to the country's GDP (Central Bureau of Statistics, 2021) (Ministry of Agricultural Development, Government of Nepal, 2015). However, traditional farming has unpredictable weather conditions and a lack of modern technology which is yet to be known, that pose challenges to productivity for farmers. Farmers still rely on conventional irrigation methods which often lead to inefficient water usage (waste of water) and crop losses. Therefore, climate change has increased the frequency of thirst on a global scale, agriculture is undergoing on a transformation with the adoption of IOT and precision farming techniques.

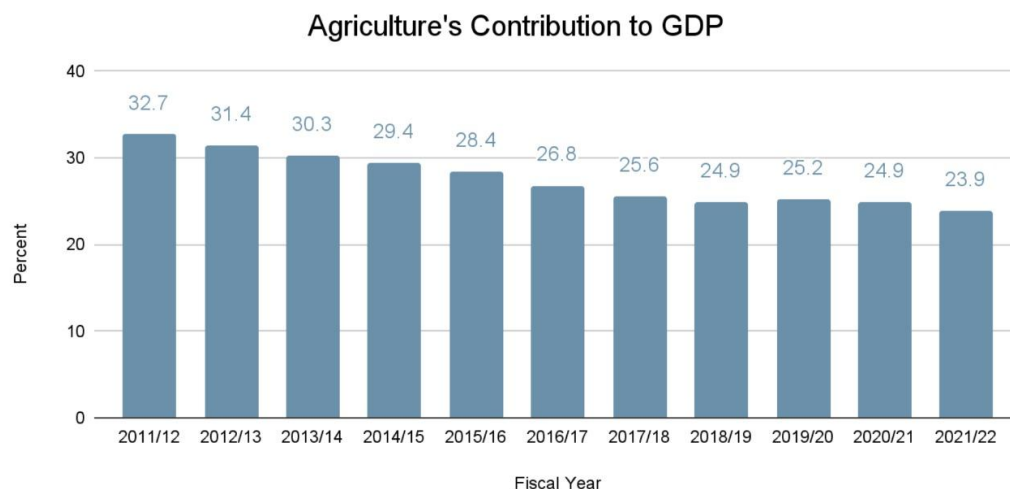


Figure 2 – Agriculture's Contribution to GDP

According to a report by Markets and Markets (2022), the global smart agriculture market is expected to grow to \$22 billion by 2025 which is a very big deal in comparison to Nepal's economy, (Nepal Rastra Bank, 2015/2022) (Anand, 2023) driven by the increasing need for sustainable farming solutions which helps the farmer grow large number of crops in a healthy way (Ministry of Agricultural Development, Government of Nepal, 2015).

## 1.2 Problem Statement

Nepal is an agriculture-based country where more than 65% of the population engages in agriculture in a daily basis for livelihood and agriculture shares about 33% of the GDP at current price and 35% at 2000/01 constant price (Ministry of Agricultural Development, Government of Nepal, 2015). Agriculture is the backbone of our country Nepal. There is traditional agriculture system in Nepal even in the twenty first century. In traditional agricultural practices farmers face a variety of challenges that reduces crop yields and sustainability including inefficient water usage (Wastage of water) poor soil quality, unpredictable weather in different areas and a lack of real time data in crop health.

### PROBLEMS AND SOLUTIONS IN AGRICULTURE

#### Major Problems in Agriculture

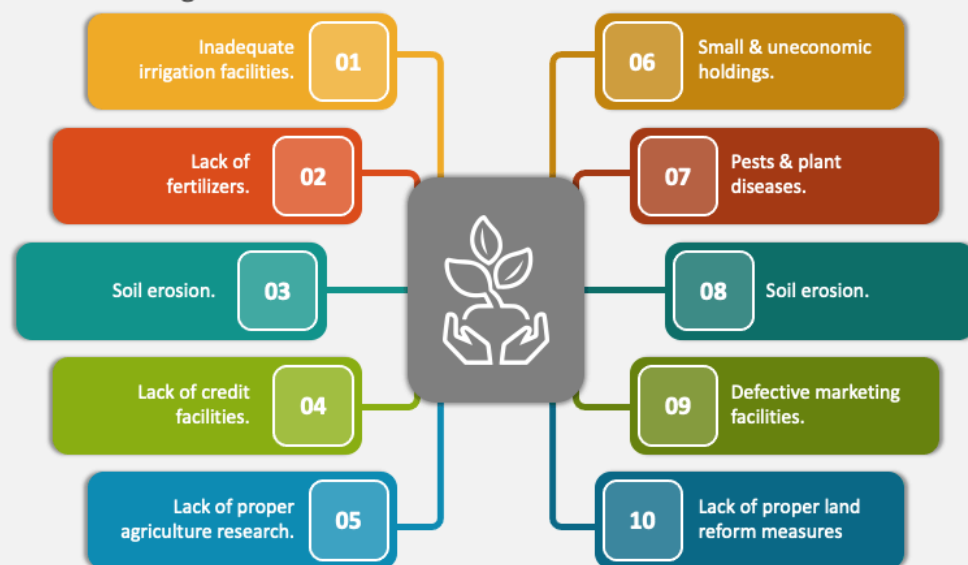


Figure 3 – Problems

Farmers sometimes provide too much water to the and affect their growth. This issue often leads to many wasted resources, increased operational costs, and lower overall productivity. Therefore, agriculture sector has not been developed satisfactorily.



### 1.3 Project as a solution

To solve these challenges, we have developed an IOT Based Smart Agricultural System farming project as a solution that uses various sensors such as ultrasonic sensor, Dht11 sensor, soil moisture sensor and devices like esp32 to optimize most of the resources. By continuously monitoring environmental conditions, the system that automatically adjust irrigation then having the rights on real time data improving crop health and enhancing overall productivity. The IOT-Based Smart Agriculture tools because it will help the farmers grow trust over the IOT-Based tools so that they know where to invest their money. A more sustainable data driven practice that maximizes resources and increases agricultural efficiency in Nepal (Sharma et al., 2022).

## 1.4 Aim and Objectives

### Aim

The main aim of developing an IOT-based smart agricultural system is to help farmer to automate agricultural task such as managing irrigation, adapting to climate change and optimizing resource management.

### Objectives

- To develop a soil moisture monitoring system that automatically activates irrigation based on real-time moisture levels ensuring crops are provided with the right amount of water preventing from over irrigation.
- To provide a user-friendly application to farmers to monitor the conditions remotely and receive alerts if any parameters go out of the expected range, potentially harming the crops.
- To monitor the water supply in storage tanks to optimize water usage and prevent shortages.
- To viral the concept of IoT Based Smart Agricultural System throughout the globe.
- To develop the country with the help of IoT devices.

## 2 Background

### 2.1 System Overview

This IOT based smart irrigation system is proposed to automate the irrigation and monitor the environmental conditions to get the optimal crop growth and use the resources minimally which will be done by integrating different sensors including ultrasonic, DHT11 (temperature, humidity) and soil moisture sensors with ESP32 microcontroller and real-time monitoring and control will be done through Blynk.

There will be soil moisture sensor implanted inside the soil which is going to get the measurements on the soil moisture level continuously and whenever the level of moisture is below the threshold it will then turn on the water pump which will irrigate the soil to needed moisture and this process will be repeated again and again so that the soil never remains dry and also there is no overflow of water ensuring the minimal use of water.

So, whenever the temperature rise or drop the system will be changing the irrigation schedule so that the crops are not overstressed with over irrigation or under irrigation also, we have the monitoring and control system whereby all the data from the sensors will be transmitted to Blynk app whereby there will be alerts, reports about conditions and from where farmers will be controlling the system remotely this is whole system with which a farmer can improve overall crop yield (Kumar and Singh, 2021).

## 2.2 Design Diagrams

### 2.2.1. Hardware Architecture:

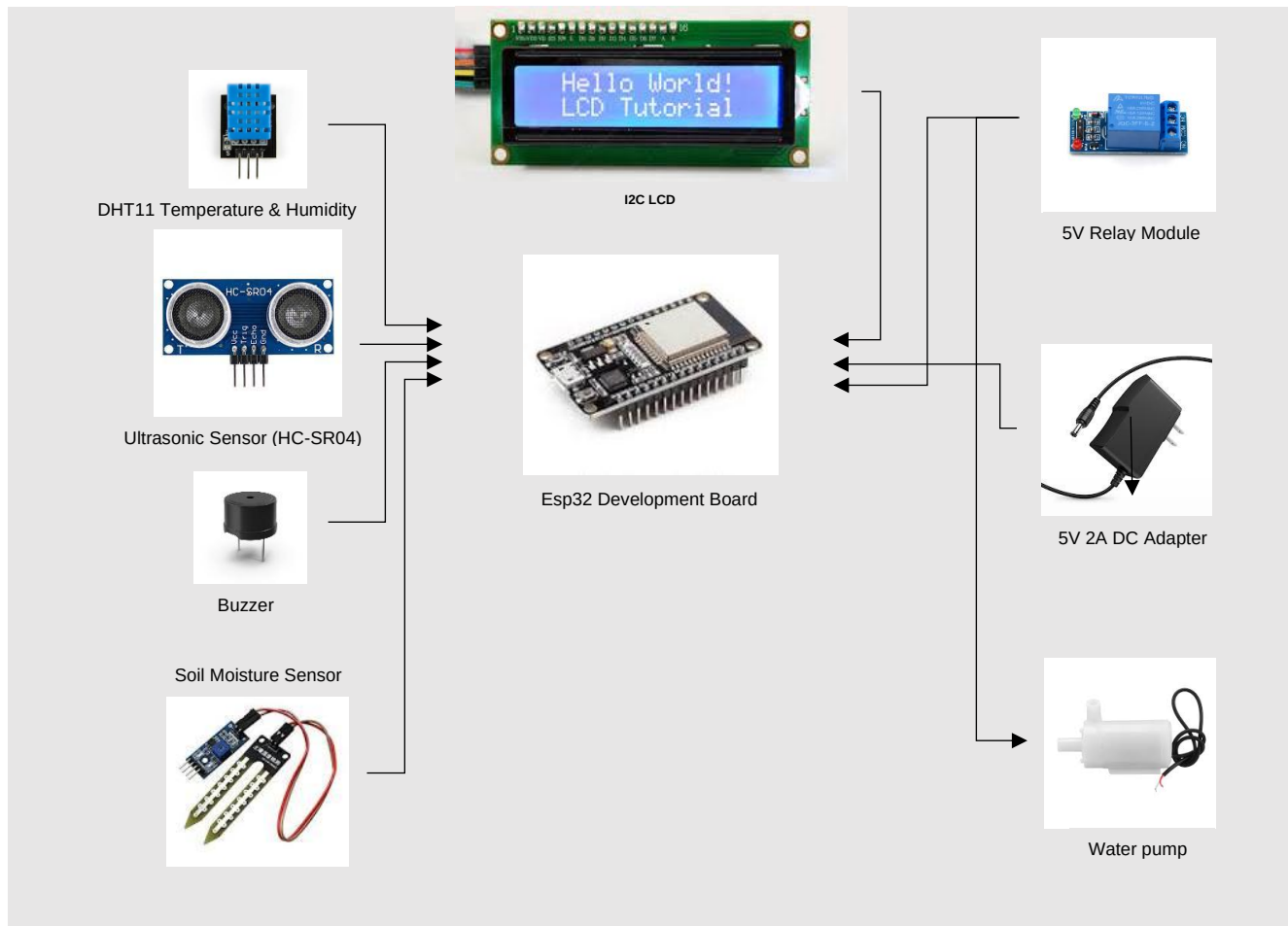


Figure 4 – Hardware Architecture

### 2.2.2. Circuit Diagram:

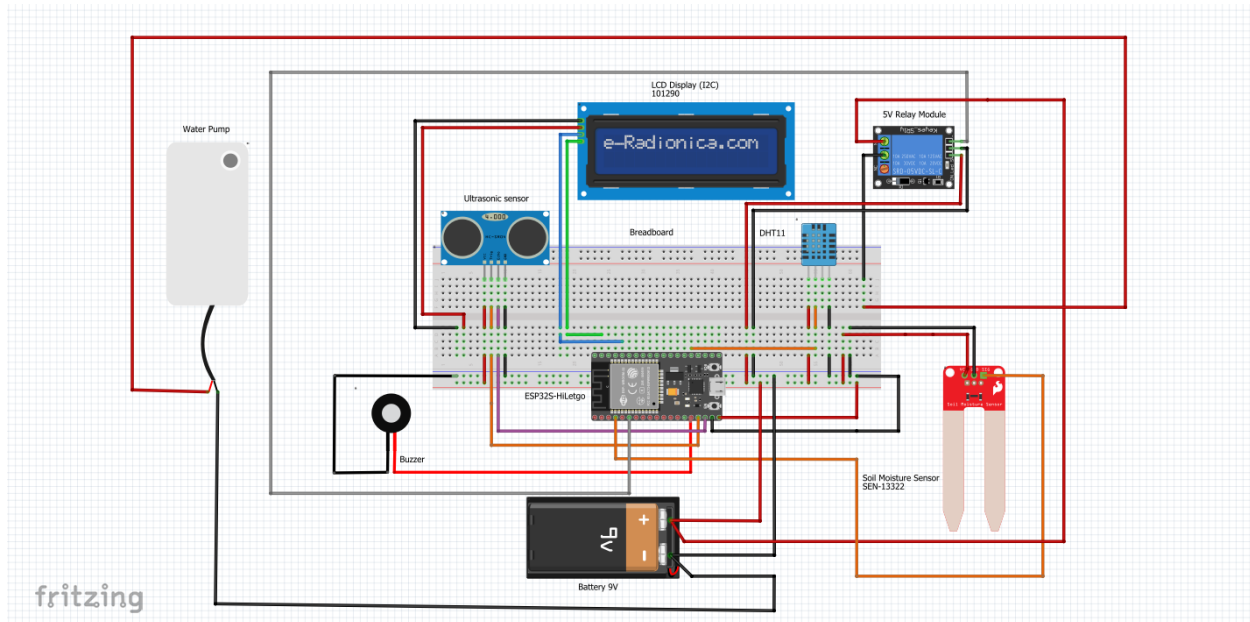


Figure 5 – Circuit Diagram

## 2.2.3. Flowchart:

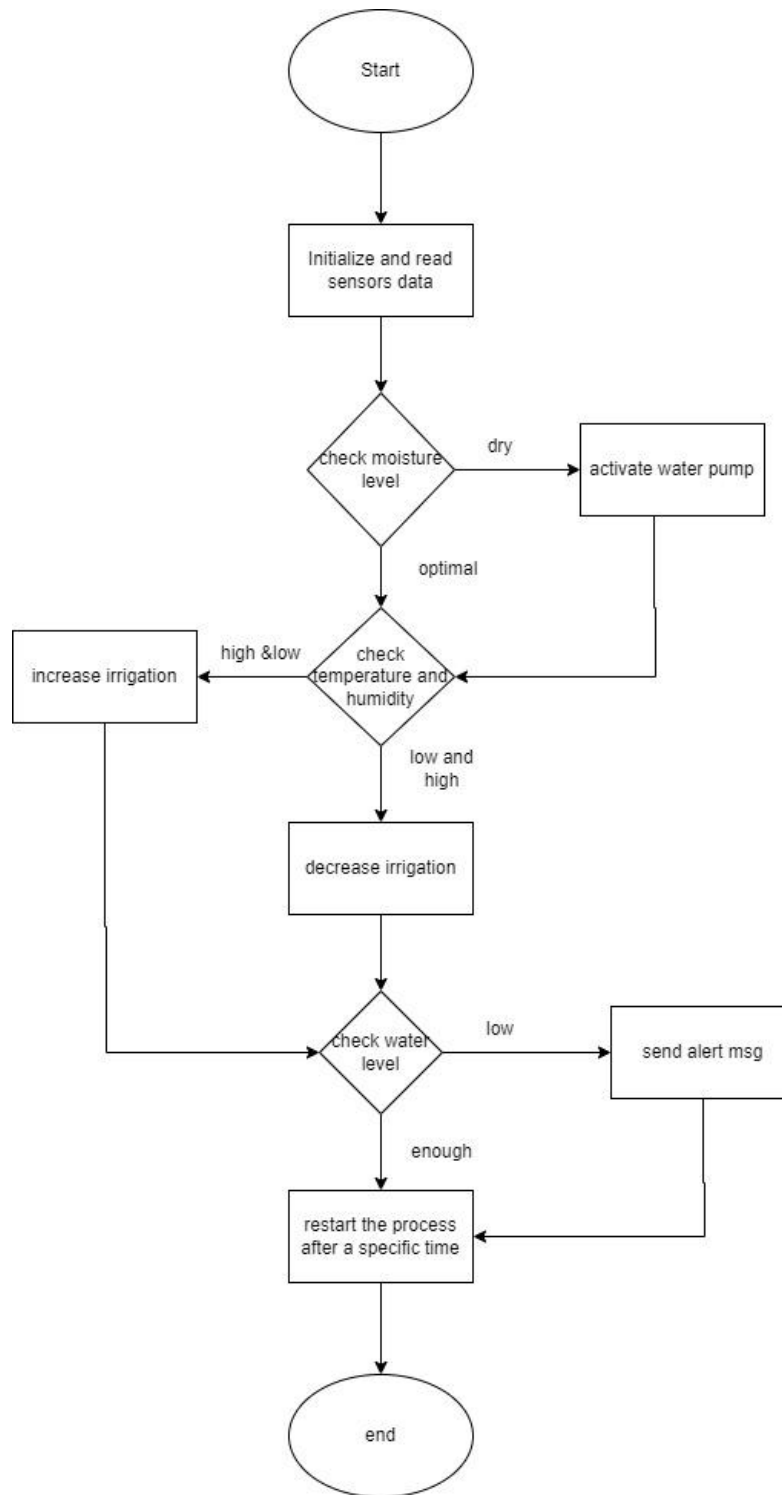


Figure 6 – Flowchart

## 2.3 Requirement Analysis

### 2.3.1 Hardware Components

The hardware required for this system are:

- **ESP32 Microcontroller:** It is a central microcontroller developed by Express if that includes Wi-Fi and Bluetooth wireless capabilities and dual-core processor used for processing and communication.
- **Soil Moisture Sensor:** It measure the soil moisture if the soil is dry it will detect and if the soil is moist when poured water it will then detect the soil is moist.
- **DHT11:** This sensor allows the user to measure the temperature and humidity.



Figure 7 – ESP-32 Microcontroller

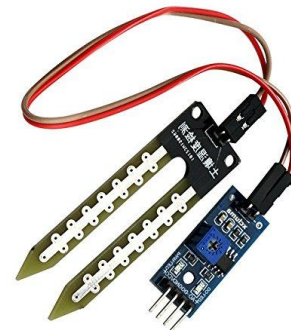


Figure 8 – Soil Moisture Sensor

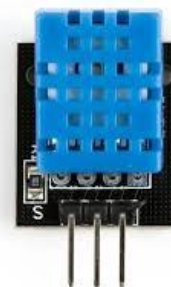


Figure 9 – DHT11 Temperature and Humidity Sensor

- **HC-SR04:** This sensor detects the distance between an object.

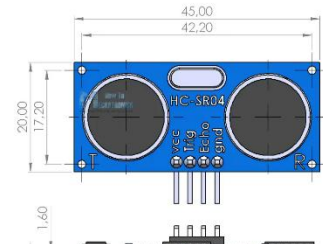


Figure 10 – Ultrasonic Sensor (HC-SR04)

- **Water pump:** It automatically irrigates the soil whenever needed by transferring the water in the tank to the soil.



Figure 11 – Water Pump

- **Buzzer:** Whenever, there is low water level or there is a bad weather it alerts the farmers with its alarm system.

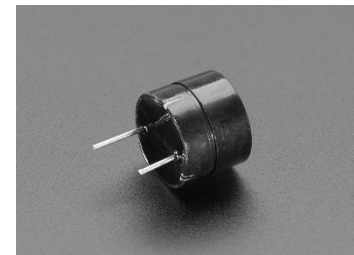


Figure 12 – Buzzer



- **5V Relay Module:** It controls the water pump based on the data provided by the sensors.

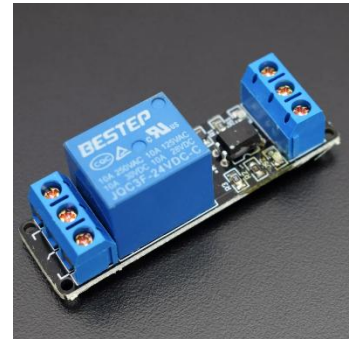


Figure 13 – 5V Relay Module

- **LCD Display:** It displays the environmental conditions.

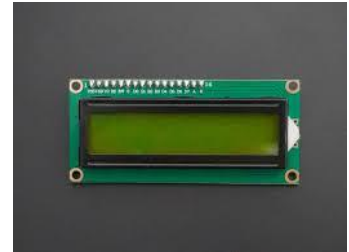


Figure 14 – I2C LCD

- **Jumper Wires:** It is used to make connection between items on your breadboard and your microcontroller header pins.

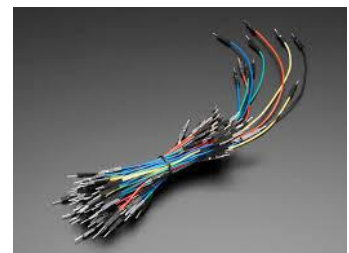


Figure 15 – Jumper Wires

- **Breadboard:** It is used for building temporary circuits.



Figure 16 – Breadboard

- **5V 2A DC Adapter:** It powers ESP32 and components.



Figure 17 – 5V 2A DC Adapter

### 2.3.2 Software Components

The software required for this system are as:

- **Arduino IDE:** All the codes for esp32 will be written over this IDE.



Figure 18 – Arduino IDE

- **Blynk:** It's a cloud platform for real-time monitoring and alerts.



### Blynk Features



Figure 19 – Blynk

- **Fritzing:** It is used for designing circuit diagrams.

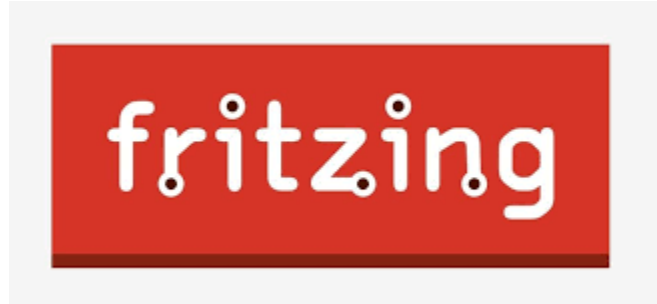


Figure 20 – Fritzing

### 3 Case Study

#### 3.1 Introduction

For this case study, we focus on a local community farm growing tomatoes, lettuce, and carrots—crops that require careful monitoring of soil moisture and climate conditions using tools like ESP32, Ultrasonic Sensor (HC-SR04), DHT11 Temperature and Humidity Sensor, Soil Moisture Sensor, Buzzer, Water Pump, LCD Display (I2C), Jumper Wires, Breadboard, 5V relay module, 5V 2A DC adapter, etc.

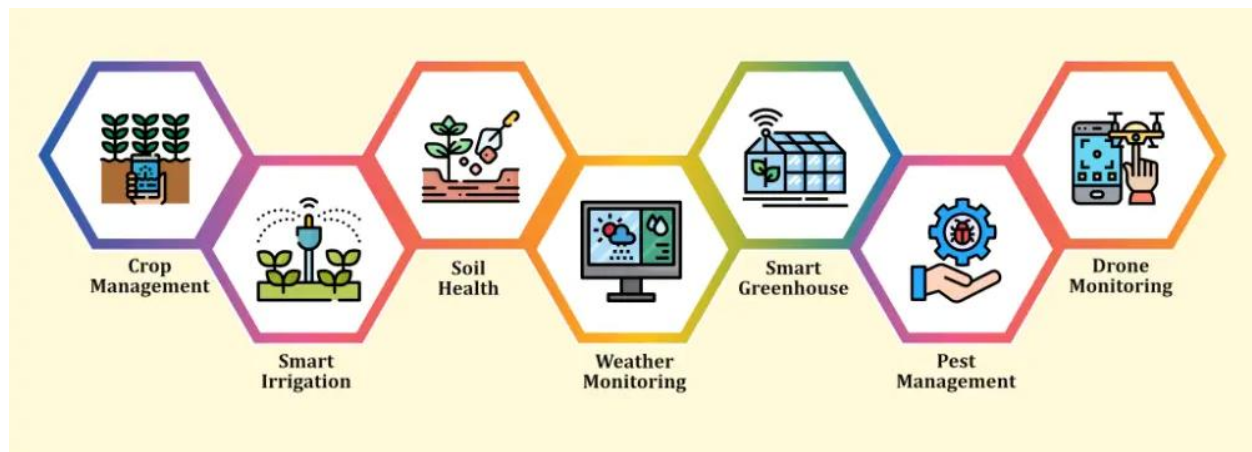


Figure 21 – Cases of Precision Farming using IoT

The implementation of an IoT-based smart agricultural system will help farmers overcome traditional inefficiencies, enabling better decision making through real time data collection and automation (Ahmed et al., 2022).

### 3.2 Problem Statement

The community farm of Nepal faces major challenges related to inconsistent irrigation, water wastage, and the impact of climate change not only that the main thing it lacks are the advanced tools of IOT system. Farmers manually check soil conditions, which often leads to overwatering or underwatering, reducing crop yields and also affects the farmer health.

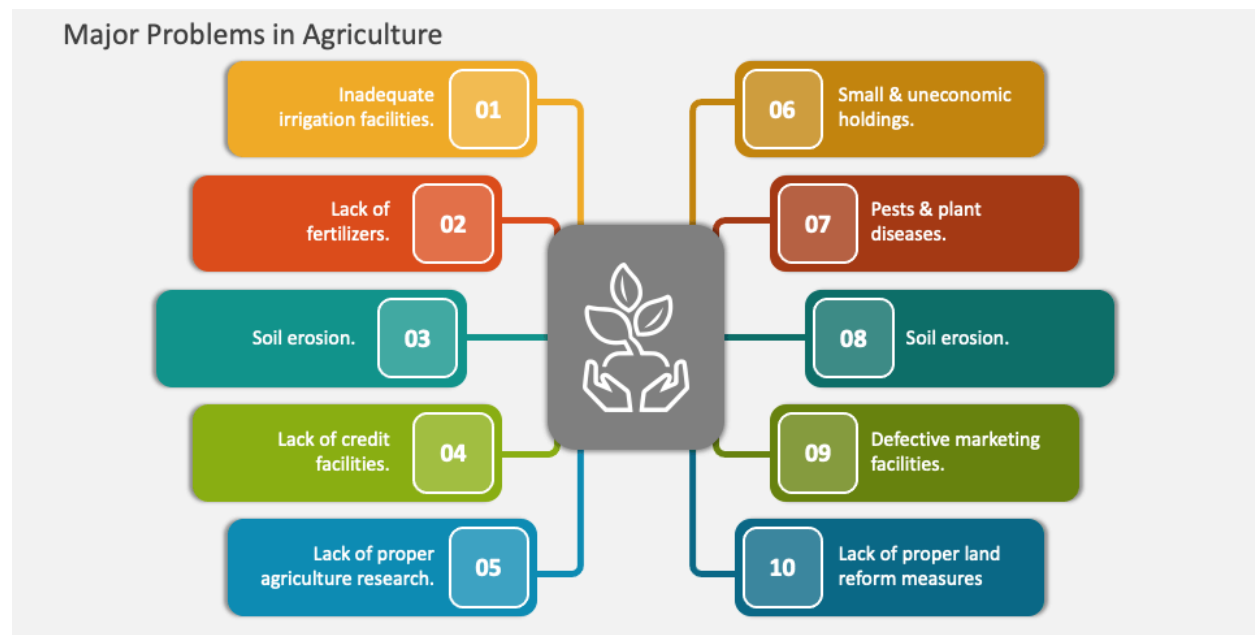


Figure 22 – Major Problems in Agriculture

Additionally, the lack of automated monitoring means farmers has limited access to real time weather or soil data and too much labor making it difficult to optimize farming practices.

### 3.3 System Analysis

By implementing an IOT (Internet of things) driven solution, the farm can improve its irrigation process by using soil moisture sensors to determine the exact water requirements of crops, applying pesticides automatically using AI-driven tools which helps to destroy insects that are harmful to the cultivated plants and humans.



Figure 23 – System Analysis

A temperature and humidity sensor will provide real-time weather updates ensuring the farm adjusts watering schedules according to environmental changes, an ultrasonic sensor will monitor water tank levels to prevent shortages and optimize resource use, Buzzer for alerts, allowing farmers to track conditions remotely and receive alerts if any parameters go beyond the optimal range (Sabu, 2023).

### 3.4 Recommendations

As from our research of this case study, we recommended that the project on the IoT based agricultural system which was invented by our team. The IoT Based system has an insane capability to develop and improve crop yields and reduce the wastage of water usage which leads to a great futuristic agricultural system and profitable for farmers as well. As from our analysis throughout this project we came to acknowledgment of all the components used synced perfectly with the system's nature and design. So, we recommend that the idea on further research should be done to explore the capability for integrating machine learning and artificial intelligence (Yadav, n.d.). By implementing this it could not only improve the agricultural system but also be used on other platforms for insane accurate predictions about crop growth and water usage.



Figure 24 – Advance IoT



Therefore, increasing the efficiency and productivity too. The IoT-Based Agricultural System has the potentiality to modernize the agricultural sector of Nepal by automation, optimization and management. This system can help farmers make good real-time decisions regarding enlarge pest control, good environment monitoring, fertilizer usage and crop harvesting times which will increase the growth of crop yields, profitable investments and cheap costs. To implement the effects of this system, it is recommended to:

- Advertise the Ideal use of Artificial Intelligence and Machine learning.
- Inspection of laws on the country before implementing the analysis.

## 4 Development

### 4.1 Planning and Designs

The phases in which we have developed an IOT-Based Smart Agricultural System using various gadgets such as: Sensors, Actuators, Microcontroller, Jumper wires, Breadboard, Adapters & cables, Water pump, Integrated circuits (Relay module), Buzzer, LCD, etc. To create a system for farmers to make there work easier and sustainable.

The designing for the IOT-Based Smart Agricultural System are given below:

- User requirements
- Aim of the project
- Developing
- Testing

## 4.2 System Development

### 4.2.1 Phase 1

#### Connecting ESP-32, Breadboard and DHT11 sensor

Initially, DHT11 was placed on breadboard. The 3.3V and GND pin of ESP32 were connected to the Breadboard positive and negative rail respectively. Then the signal pin of DHT11 was connected to GPIO5 on ESP32, while the DHT VCC and GND of the DHT11 were connected to the breadboard positive and negative port.

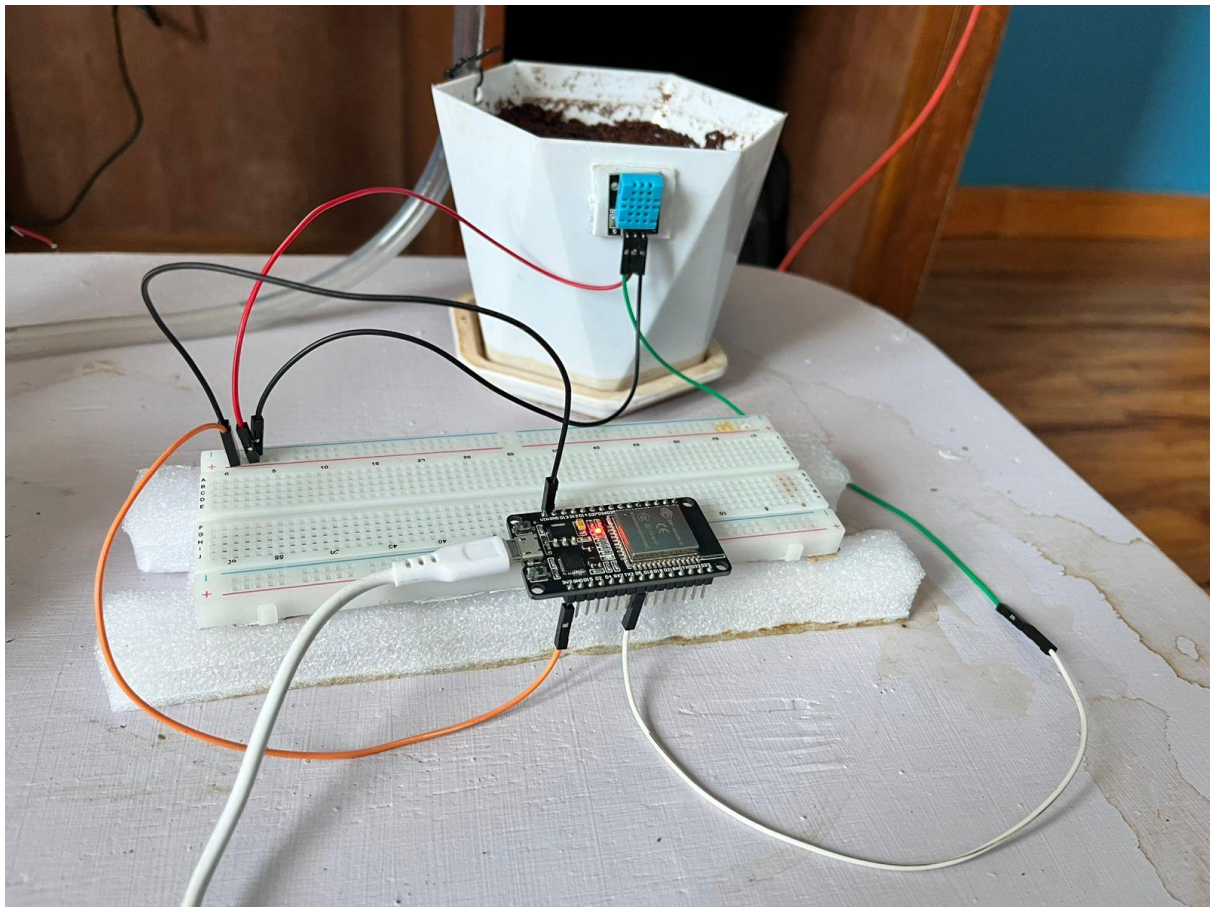


Figure 25 – Phase 1\_Connecting DHT11, Breadboard and ESP-32

### 4.2.2 Phase 2

#### Connecting the Ultrasonic Sensor (HC-SR04) with the Circuits

Moving on ultrasonic sensor, Trig and Echo pin of ultrasonic sensor (HC-SR04) were connected to GPIO12 and GPIO13 of ESP32. Further, ultrasonic VCC and GND were connected to the breadboard positive and negative port respectively.

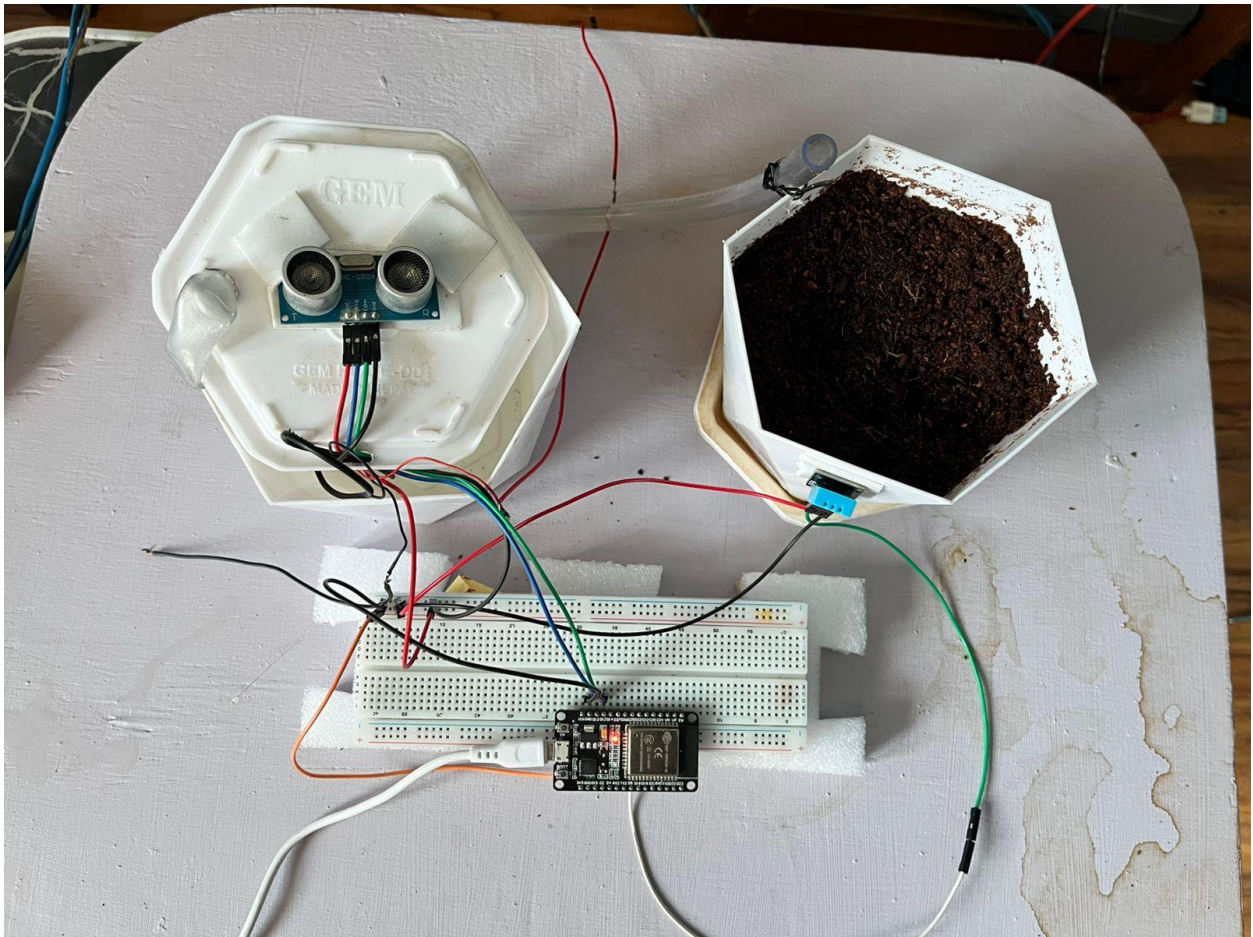


Figure 26 – Phase 2\_Connecting Ultrasonic sensor with the rest of the circuits



### 4.2.3 Phase 3

#### Connecting Buzzer with Soil moisture sensor

In third phase, buzzer positive terminal was connected to ESP32 GPIO14 pin and buzzer negative terminal was connected to breadboard negative port respectively. Similarly, Soil moisture sensor positive and negative were connected to breadboard positive and negative. Analog pin(A0) of soil moisture sensor was then connected to the GPIO34 pin of esp32.

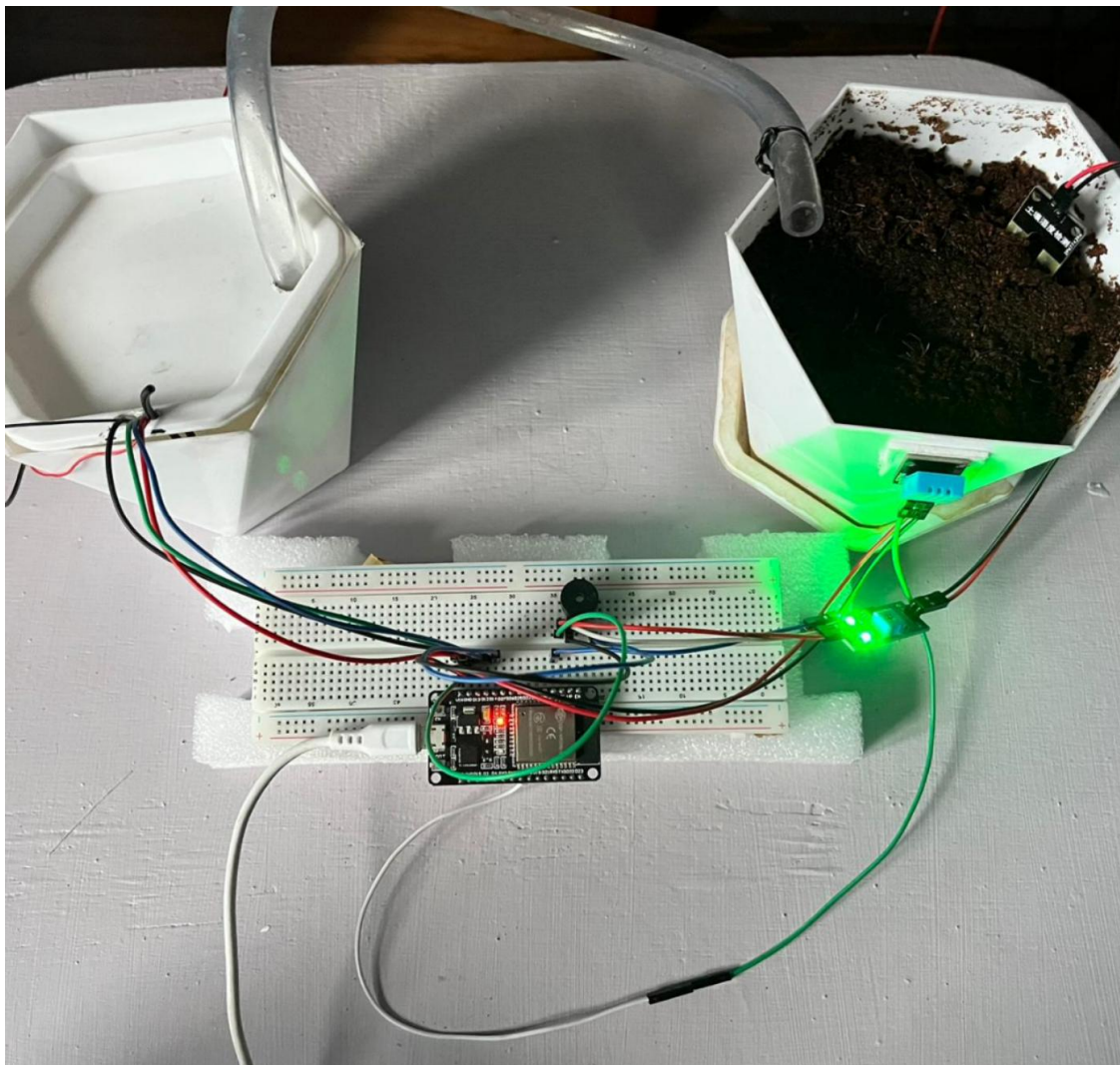


Figure 27 – Phase 3\_Connecting buzzer and soil moisture sensor with the rest of the circuits

#### 4.2.4 Phase 4

##### Connecting I2C LCD with ESP-32

In the fourth phase, I2C LCD was added with the rest of the circuits. I2C LCD VCC and GND were connected to breadboard positive and negative port. In other hand, I2C LCD SDA (Serial Data Line) and SCL (Serial Clock Line) were connected to the ESP32 GPIO21 and GPIO22 respectively.

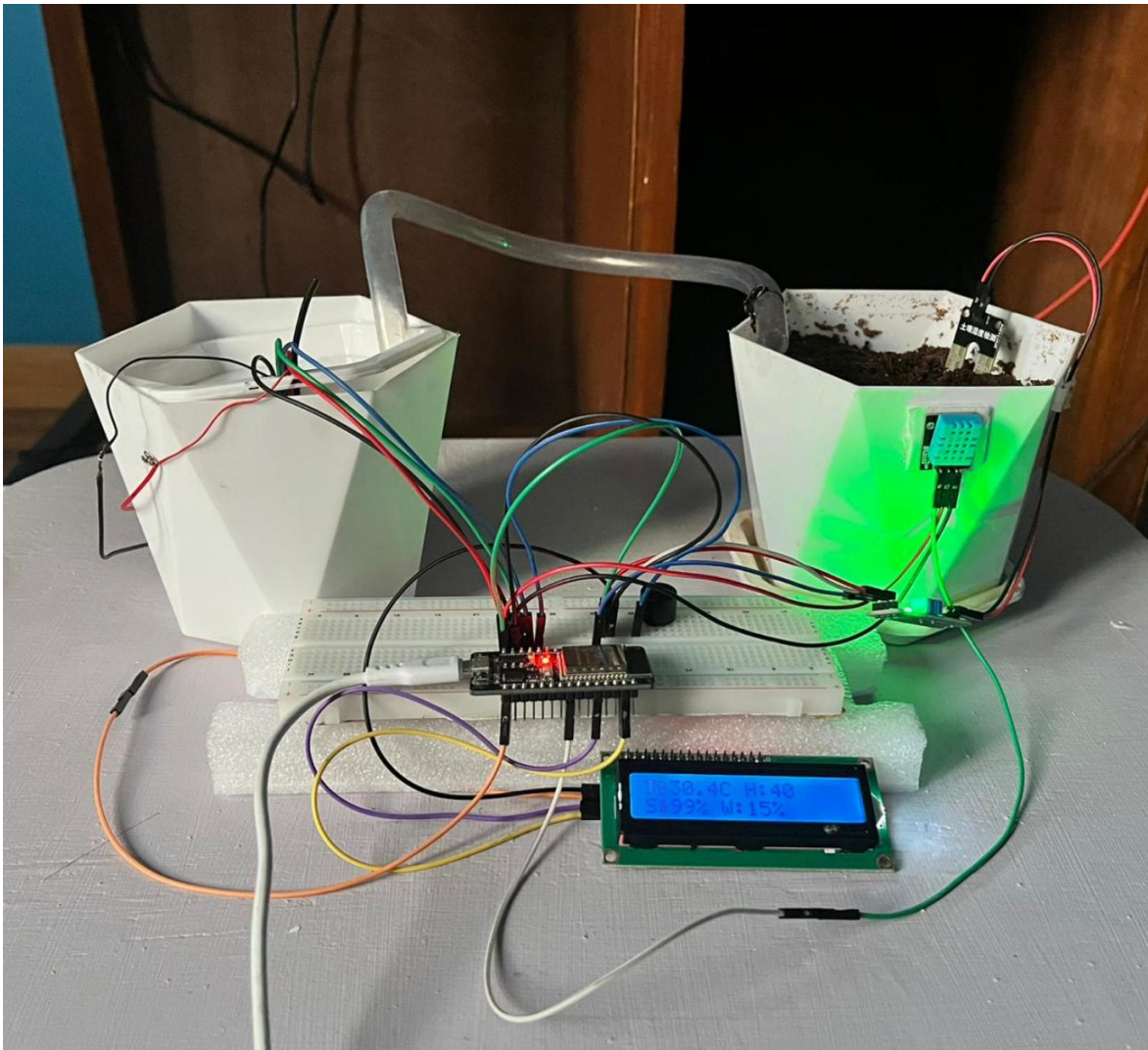


Figure 28 – Phase 4\_Connecting I2C LCD with the rest of the circuits

#### 4.2.5 Phase 5

##### Connecting 5V Relay Module, Water Pump and 5V 2A DC Adapter

In Last phase, 5V Relay Module, Water Pump and 5V 2A DC was added with the whole circuit. Relay VCC and GND were connected to breadboard positive and negative port respectively. In other hand, Relay IN pin was connected to ESP32 GPIO32 pin. Then, positive of adapter were connected to NO (Normally Open) port of 5V Relay Module and negative of both water pump and adapter were connected to the negative of breadboard. Lastly, Water pump positive pin was connected to 5V Relay Module COM(Common).

Thus, after connecting all the hardware component and uploading the code, the system functioned successfully.



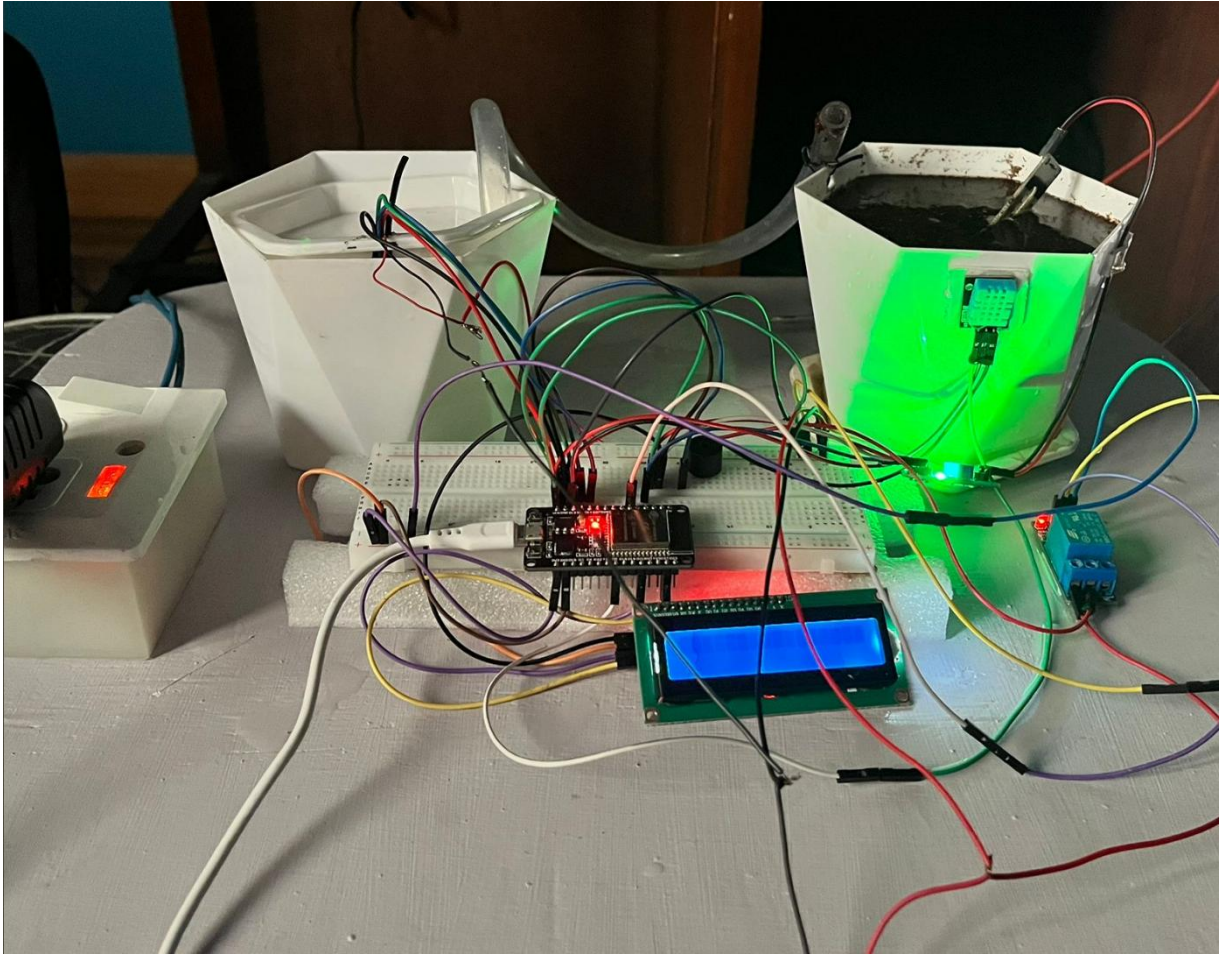


Figure 29 – Phase 5\_Connecting 3V relay module, Water pump and 5V 2A DC Adapter with the rest of the circuits



## 4.2.6 Phase 6

### Connecting with Blynk

In last phase, the whole circuit was connected with blynk to monitor it through online.

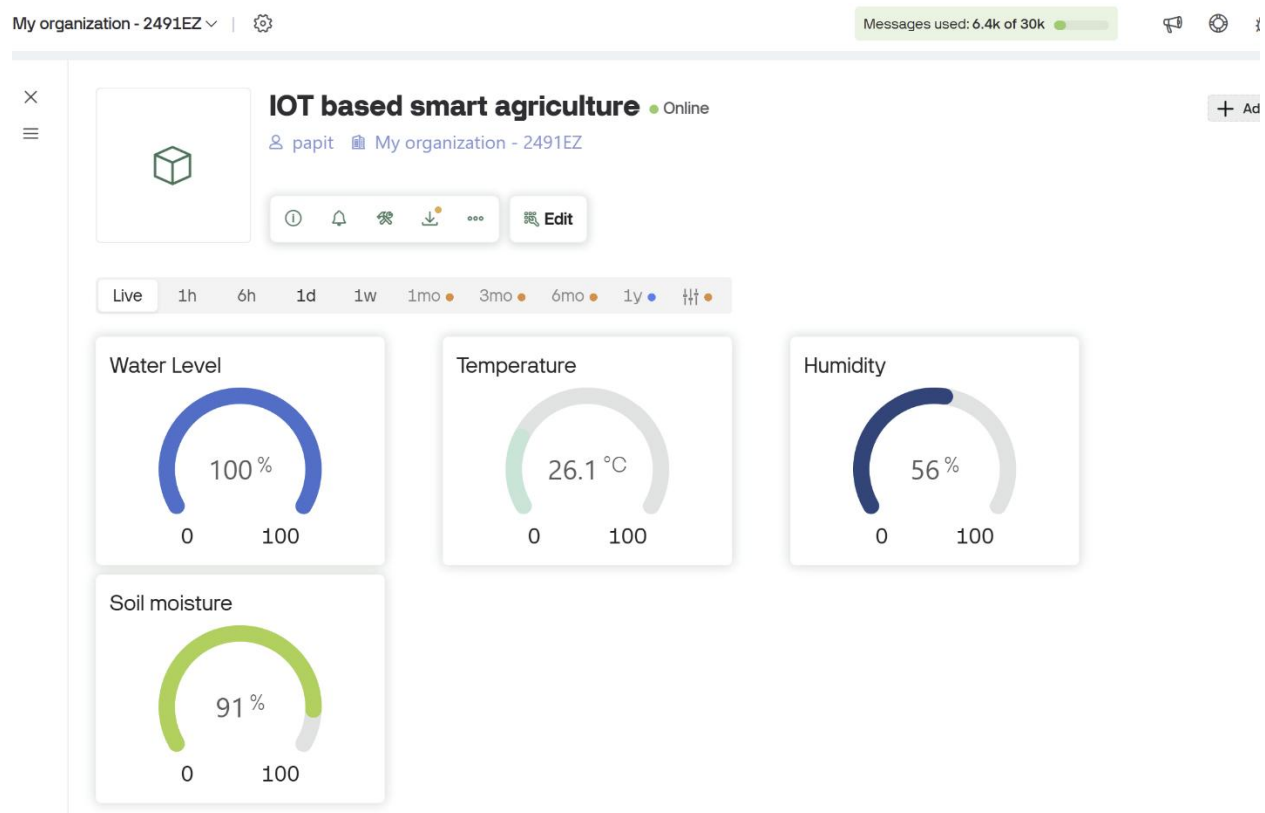


Figure 30 – Phase 6\_Conneting Blynk

### 4.2.7 Pin Connections among the Hardware Components

Table 1 – Pin Connections

| Pin Name                                  | ESP32 Pin | Breadboard | 5V Relay Module |
|-------------------------------------------|-----------|------------|-----------------|
| DHT11 - Signal Pin                        | GPIO5     | -          | -               |
| DHT11 - GND                               | -         | Negative   | -               |
| DHT11 - VCC                               | -         | Positive   | -               |
| Ultrasonic Sensor (HC-SR04)<br>- Trig Pin | GPIO12    | -          | -               |
| Ultrasonic Sensor (HC-SR04)<br>- Echo Pin | GPIO13    | -          | -               |
| Ultrasonic Sensor (HC-SR04)<br>- VCC      | -         | Positive   | -               |
| Ultrasonic Sensor (HC-SR04)<br>- GND      | -         | Negative   | -               |
| I2C LCD - SDA                             | GPIO 21   | -          | -               |
| I2C LCD - SCL                             | GPIO 22   | -          | -               |
| I2C LCD - VCC                             | -         | Positive   | -               |
| I2C LCD - GND                             | -         | Negative   | -               |
| Soil Moisture Sensor - VCC                | -         | Positive   | -               |
| Soil Moisture sensor - GND                | -         | Negative   | -               |
| Soil Moisture Sensor - A0                 | GPIO34    | -          | -               |

|                       |        |          |     |
|-----------------------|--------|----------|-----|
| Buzzer - Positive     | GPIO14 | -        | -   |
| Buzzer - Negative     | -      | Negative | -   |
| 5V Relay Module – VCC | -      | Positive | -   |
| 5V Relay Module – GND | -      | Negative | -   |
| 5V Relay Module – IN  | -      | GPIO32   | -   |
| Adapter - Positive    | -      | Positive | NO  |
| Adapter – Negative    | -      | Negative | -   |
| Water Pump – Positive | -      | -        | COM |
| Water Pump – Negative | -      | Negative | -   |

## 5 Results and Findings

### 5.1 Result

After integrating all the sensors and actuators. The advanced IOT Based agriculture system was able to successfully monitor and respond to our implementations. The DHT11 sensor worked well and tracked temperature and humidity levels, while the soil moisture sensors tracked consistent real-time data on soil conditions. These readings were then correctly displayed on the I2C LCD screen making it easy to monitor the environment of the system. The use of the ultrasonic sensor allowed for basic distance measurements acknowledging the setup could measure the distance of the water control efficiently.

The 5V Relay module in connection with a mini water pump functioned reliably whenever the system detected low moisture of the soil (dry soil). This automatic irrigation triggered only when necessary, preventing wastage of water. Moreover, the work of a buzzer helped with alerts for specific conditions like high humidity (DHT11 sensor) or low moisture (Soil moisture sensor) making it more interactive and practical. One key finding was that the system, despite being budget-friendly and compact, delivered stability and gave accurate outputs over multiple test runs. However, during testing the microcontroller needed reboot time to time.

Overall, the project met its main objective on making the work of the farmers by reducing manual labor in farming through automation using IOT. The combination of hardware and software worked well in tandem to simulate a smart mini agriculture environment. While the system is scalable improvements such as remote monitoring via Wi-Fi or GSM, real-time cloud logging, or solar power integration can further make the project suitable for larger-scale farming needs.

## 5.2 Testing

### 5.2.1 Test 1:

Table 2 – Test 1

|                        |                                                                                                                           |
|------------------------|---------------------------------------------------------------------------------------------------------------------------|
| <b>Test</b>            | 1                                                                                                                         |
| <b>Objective</b>       | To test the working of DHT11 sensor.                                                                                      |
| <b>Activity</b>        | Power was supplied to the DHT11 sensor and the temperature and humidity readings were taken at different intervals.       |
| <b>Expected Result</b> | The sensor should provide valid readings with the temperature and humidity values fluctuating as the environment changes. |
| <b>Actual Result</b>   | The sensor provided accurate readings.                                                                                    |
| <b>Conclusion</b>      | Test was successful                                                                                                       |

## Output

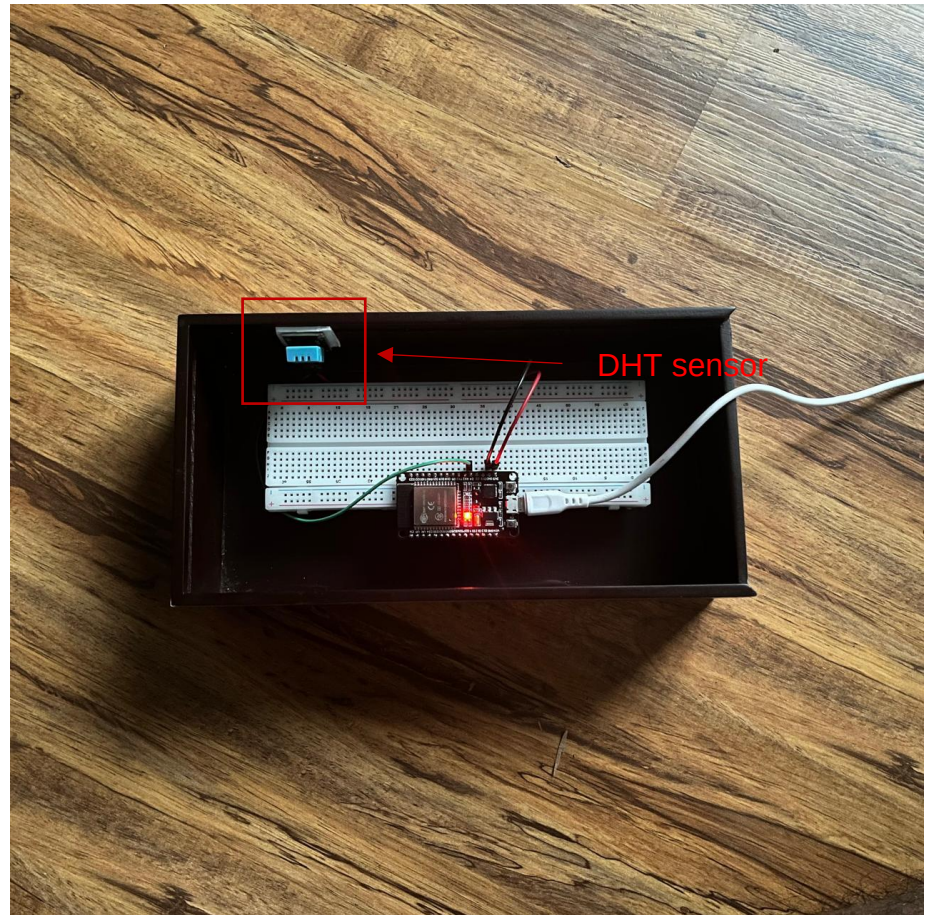


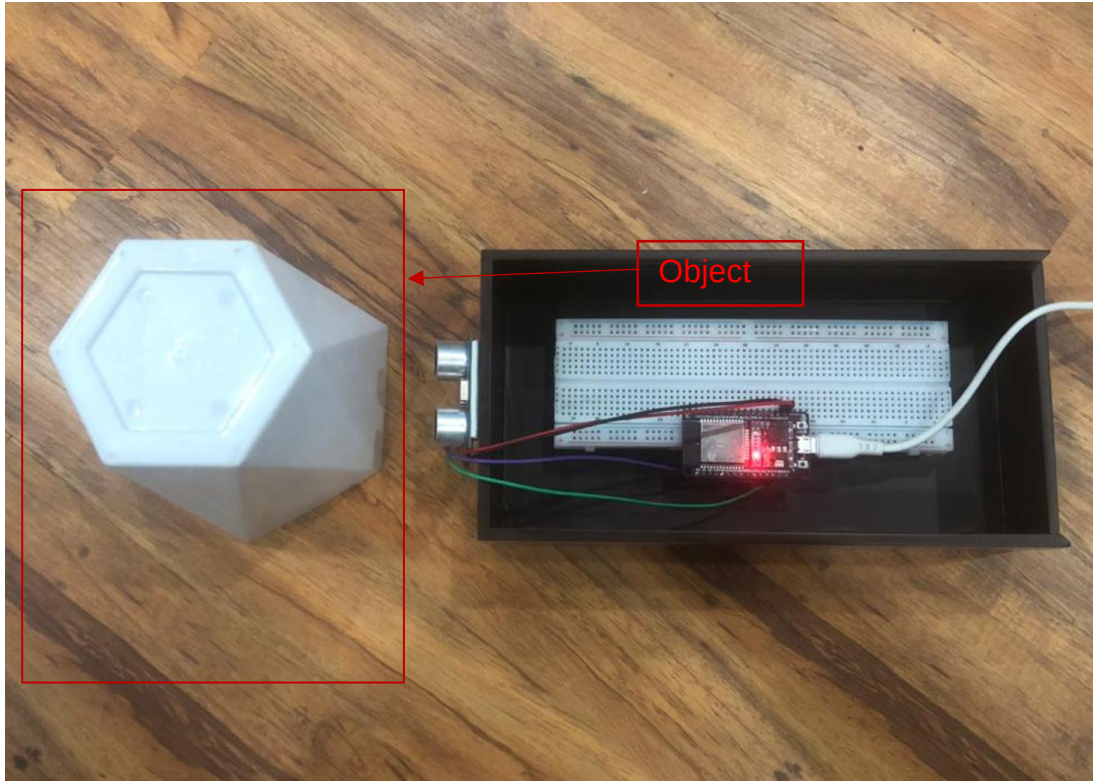
Figure 31 – Test 1\_Connecting DHT sensor with ESP-32

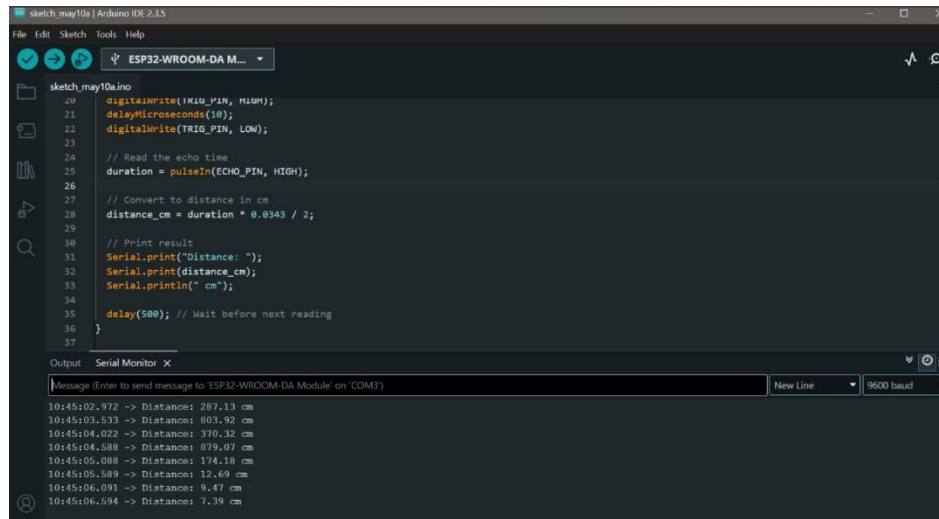
```
File Edit Sketch Tools Help
ESP32-WROOM-DA M...
sketch_apr5aino
27 if (isnan(humidity)) {
28   Serial.println("Failed to read humidity!");
29 }
30 if (isnan(temperature)) {
31   Serial.println("Failed to read temperature!");
32 }
33
34 if (isnan(humidity) || isnan(temperature)) {
35   Serial.println("Failed to read from DHT sensor!");
36   return;
37 }
38
39 // Clear the previous data in the Serial Monitor
40 Serial.println();
41
42 // Print the results to the Serial Monitor
43 Serial.print("Humidity: ");
44 Serial.print(humidity);
45 Serial.print(" %\n");
46 Serial.print("Temperature: ");
47 Serial.print(temperature);
48 Serial.print(" °C\n");
49 }
50
Output Serial Monitor X
Message (Enter to send message to 'ESP32-WROOM-DA Module' on 'COM3')
21:25:17.908 -> Humidity: 39.00 % Temperature: 25.70 °C
21:25:20.889 -> Humidity: 46.00 % Temperature: 25.70 °C
21:25:23.914 -> Humidity: 57.00 % Temperature: 25.70 °C
21:25:26.972 -> Humidity: 63.00 % Temperature: 25.80 °C
21:25:29.983 -> Humidity: 65.00 % Temperature: 26.00 °C
21:25:33.026 -> Humidity: 71.00 % Temperature: 26.10 °C
21:25:36.049 ->
```

Figure 32 – Test 1\_Testing DHT sensor with ESP-32

## 5.2.2 Test 2:

Table 3 – Test 2

|                        |                                                                                                                                                                         |
|------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Test</b>            | 2                                                                                                                                                                       |
| <b>Objective</b>       | To test the working of Ultrasonic Sensor (HC-SR04).                                                                                                                     |
| <b>Activity</b>        | Power was supplied to the ultrasonic sensor, distance readings were monitored.                                                                                          |
| <b>Expected Result</b> | The ultrasonic sensor (HC-SR04) should accurately detect and display distance measurement based on object position from it.                                             |
| <b>Actual Result</b>   | Ultrasonic sensor accurately detected and displayed measurement corresponding to object position.                                                                       |
| <b>Conclusion</b>      | Test was successful!                                                                                                                                                    |
| <b>Output</b>          |  <p>Figure 33 – Test 2_Conncting Ultrasonic sensor with the rest of the circuits</p> |



```
sketch_may10a.ino
20 digitalWrite(TRIG_PIN, HIGH);
21 delayMicroseconds(10);
22 digitalWrite(TRIG_PIN, LOW);
23
24 // Read the echo time
25 duration = pulseIn(ECHO_PIN, HIGH);
26
27 // Convert to distance in cm
28 distance_cm = duration * 0.0343 / 2;
29
30 // Print result
31 Serial.print("Distance: ");
32 Serial.print(distance_cm);
33 Serial.println(" cm");
34
35 delay(500); // Wait before next reading
36
37
```

Output Serial Monitor X

Message (Enter to send message to 'ESP32-WROOM-DA Module' on 'COM3')

New Line 9600 baud

10:45:02.972 -> Distanon: 287.13 cm  
10:45:03.533 -> Distanon: 803.92 cm  
10:45:04.022 -> Distanon: 370.32 cm  
10:45:04.588 -> Distanon: 879.07 cm  
10:45:05.088 -> Distanon: 174.18 cm  
10:45:05.589 -> Distanon: 12.69 cm  
10:45:06.091 -> Distanon: 9.47 cm  
10:45:06.594 -> Distanon: 7.39 cm

Figure 34 – Test 2\_Testing Ultrasonic sensor with the rest of the circuits



## 5.2.3 Test 3:

Table 4 – Test 3

|                        |                                                                                                                                                                                |
|------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Test</b>            | 3                                                                                                                                                                              |
| <b>Objective</b>       | To test the working of soil moisture and buzzer.                                                                                                                               |
| <b>Activity</b>        | Soil moisture sensor was connected to the ESP32, and the analog value was read and printed on the serial monitor and alert buzzer when soil moisture is low.                   |
| <b>Expected Result</b> | The soil moisture sensor should display values on the serial monitor with the readings changing according to the soil's moisture level and alert user with the help of buzzer. |
| <b>Actual Result</b>   | The soil moisture sensor successfully output values on the serial monitor and alert when soil moisture is low using buzzer.                                                    |
| <b>Conclusion</b>      | Test was successful!                                                                                                                                                           |

## Output

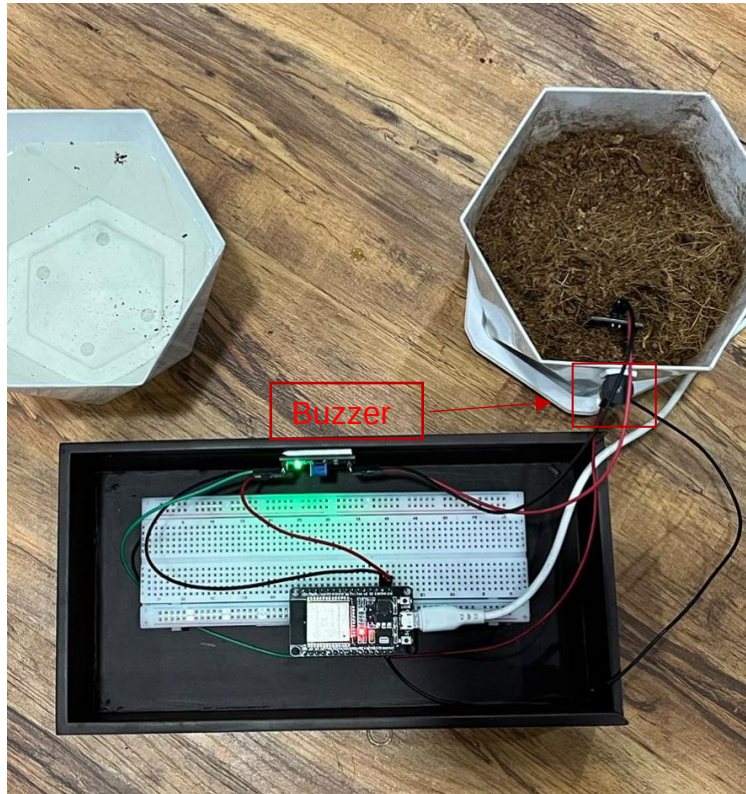
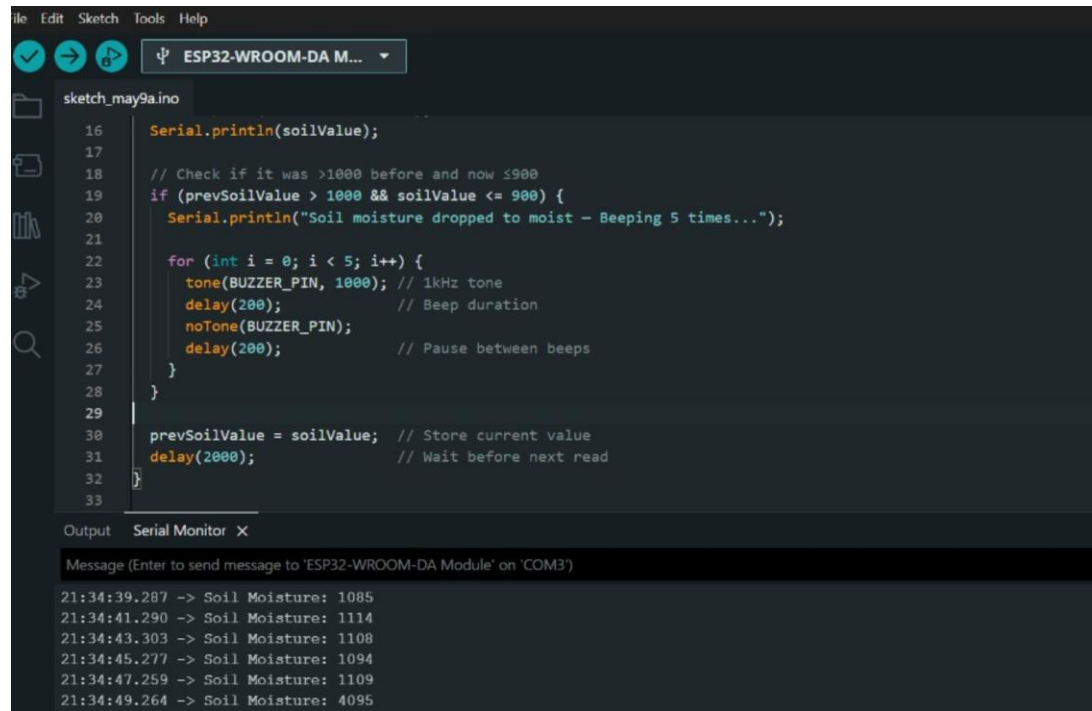


Figure 35 – Test 3\_Connecting Buzzer and Soil moisture sensor with the rest of the circuits



```
file Edit Sketch Tools Help
ESP32-WROOM-DA M...
sketch_may9a.ino
16 Serial.println(soilValue);
17
18 // Check if it was >1000 before and now <900
19 if (prevSoilValue > 1000 && soilValue <= 900) {
20   Serial.println("Soil moisture dropped to moist - Beeping 5 times...");
21
22   for (int i = 0; i < 5; i++) {
23     tone(BUZZER_PIN, 1000); // 1kHz tone
24     delay(200);             // Beep duration
25     noTone(BUZZER_PIN);
26     delay(200);             // Pause between beeps
27   }
28 }
29
30 prevSoilValue = soilValue; // Store current value
31 delay(2000);              // Wait before next read
32
33
Output Serial Monitor X
Message (Enter to send message to 'ESP32-WROOM-DA Module' on 'COM3')
21:34:39.287 -> Soil Moisture: 1085
21:34:41.290 -> Soil Moisture: 1114
21:34:43.303 -> Soil Moisture: 1108
21:34:45.277 -> Soil Moisture: 1094
21:34:47.259 -> Soil Moisture: 1109
21:34:49.264 -> Soil Moisture: 4095
```

Figure 36 – Test 3\_Testing Buzzer and Soil moisture sensor with the rest of the circuits

## 5.2.4 Test 4:

Table 5 – Test 4

|                        |                                                        |
|------------------------|--------------------------------------------------------|
| <b>Test</b>            | 4                                                      |
| <b>Objective</b>       | To test the printing functionality of the LCD display. |
| <b>Activity</b>        | Uploaded the code to Arduino using Arduino IDE.        |
| <b>Expected Result</b> | It should display “Hello World” message.               |
| <b>Actual Result</b>   | It displayed “Hello World”.                            |
| <b>Conclusion</b>      | Test was successful!                                   |

## Output

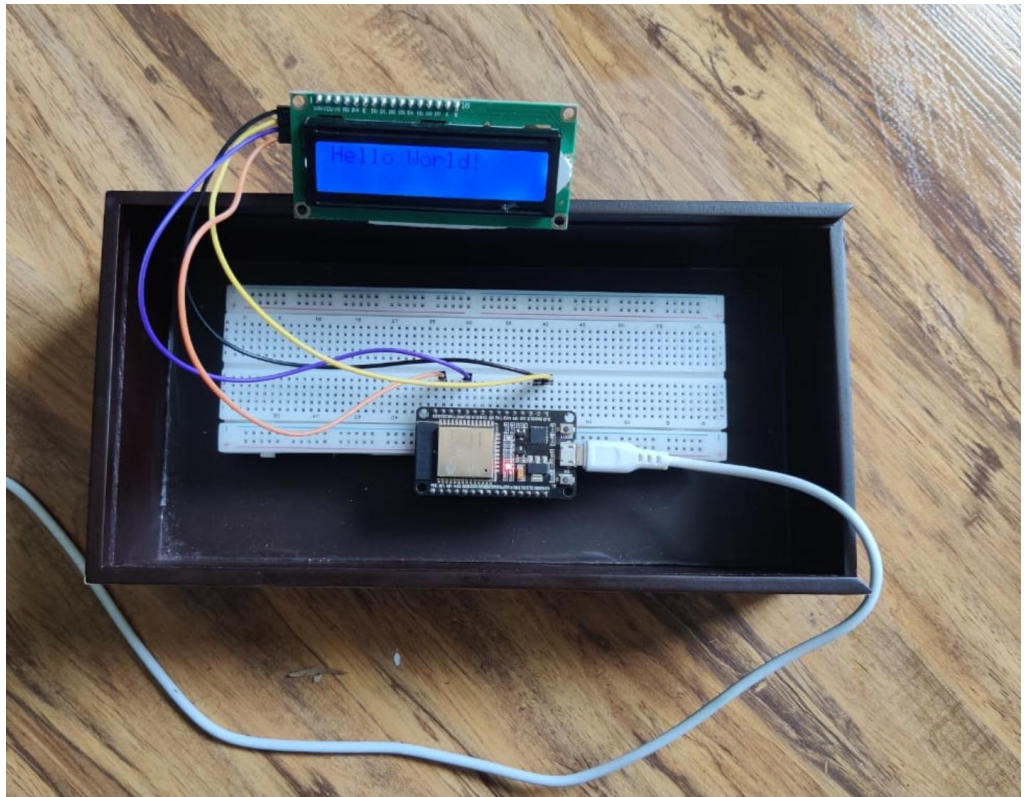


Figure 37 – Test 4\_Connecting I2C LCD with the rest of the circuits

```
File Edit Sketch Tools Help
ESP32 Dev Module

Blynkino
1 #include <Wire.h>
2 #include <LiquidCrystal_I2C.h>
3
4 // Set the LCD I2C address - 0x27 is common, change it if yours is different
5 #define I2C_ADDR 0x27
6 #define LCD_COLUMNS 16
7 #define LCD_ROWS 2
8
9 // Initialize the LCD
10 LiquidCrystal_I2C lcd(I2C_ADDR, LCD_COLUMNS, LCD_ROWS);
11
12 void setup() {
13   // Initialize I2C and LCD
14   lcd.init();
15   lcd.backlight(); // Turn on backlight
16
17   lcd.setCursor(0, 0); // Column 0, Row 0
18   lcd.print("Hello, World!");
19 }

Output
WARNING: library LiquidCrystal_I2C claims to run on avr architecture(s) and may be incompatible with your current board which runs on esp32 architecture(s).
Sketch uses 309242 bytes (23%) of program storage space. Maximum is 1310720 bytes.
Global variables use 21836 bytes (6%) of dynamic memory, leaving 305844 bytes for local variables. Maximum is 327680 bytes.

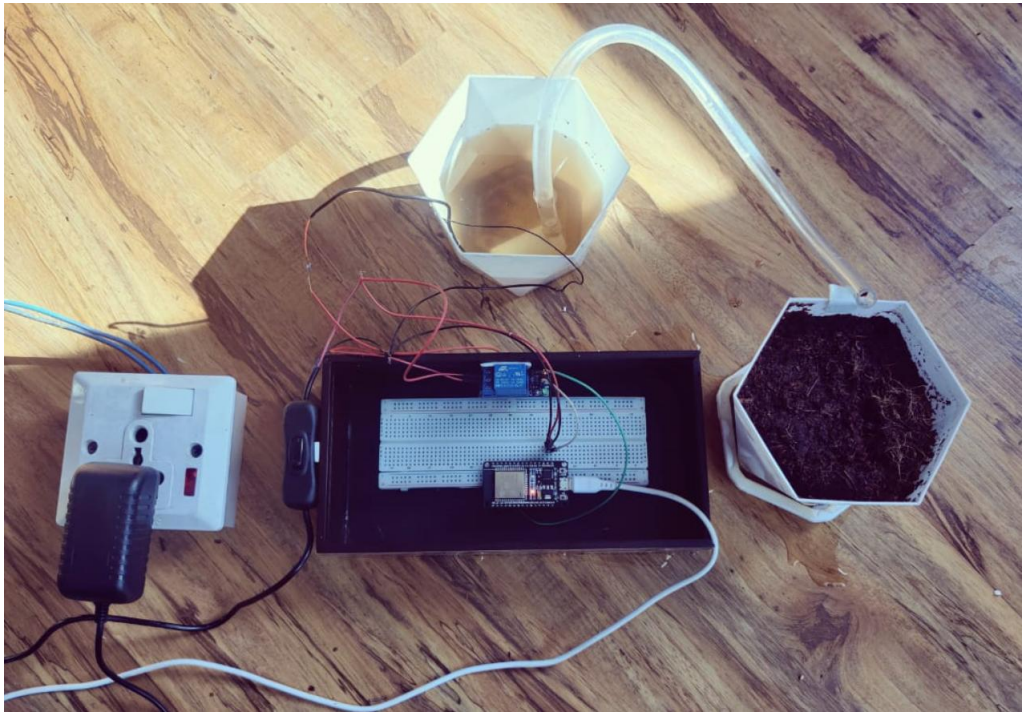
Ln 4, Col 15 ESP32 Dev Module [not connected]
```

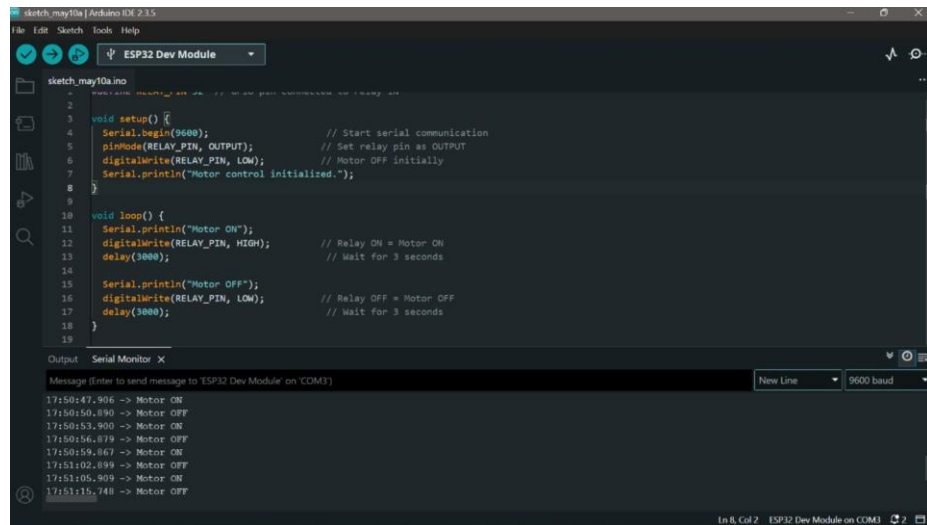
Figure 38 – Test 4\_Testing I2C LCD with the rest of the circuits



## 5.2.5 Test 5:

Table 6 – Test 5

|                        |                                                                                                                                                                                                       |
|------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Test</b>            | 5                                                                                                                                                                                                     |
| <b>Objective</b>       | To test the working of Adapter, Water pump and 5V Relay Module                                                                                                                                        |
| <b>Activity</b>        | The water pump, relay, and adapter were connected to the ESP32. The relay was controlled through the GPIO32 pin and the water pump was turned on and off by toggling the relay.                       |
| <b>Expected Result</b> | The water pump should turn on when relay is activated and the water pump should turn off when relay is deactivated.                                                                                   |
| <b>Actual Result</b>   | The relay successfully activated and deactivated the water pump when the control pin on the ESP32 was toggled between LOW and HIGH                                                                    |
| <b>Conclusion</b>      | Test was successful!                                                                                                                                                                                  |
| <b>Output</b>          |  <p>Figure 39 – Test 5_Connecting 3V Relay module, Water pump and 5V DC Adapter with the rest of the circuits</p> |



```
sketch_may10a.ino
1 // Test 5_ Testing 3V Relay module, Water pump and 5V DC Adapter
2 // with the rest of the circuits
3
4 void setup() {
5   Serial.begin(9600);           // Start serial communication
6   pinMode(RELAY_PIN, OUTPUT);  // Set relay pin as OUTPUT
7   digitalWrite(RELAY_PIN, LOW); // Motor OFF initially
8   Serial.println("Motor control initialized.");
9 }
10
11 void loop() {
12   Serial.println("Motor ON");
13   digitalWrite(RELAY_PIN, HIGH); // Relay ON = Motor ON
14   delay(3000);                   // Wait for 3 seconds
15   Serial.println("Motor OFF");
16   digitalWrite(RELAY_PIN, LOW);  // Relay OFF = Motor OFF
17   delay(3000);                   // Wait for 3 seconds
18 }
19
```

Output Serial Monitor X

Message (Enter to send message to 'ESP32 Dev Module' on COM3)

17:50:47.906 -> Motor ON  
17:50:50.890 -> Motor OFF  
17:50:53.900 -> Motor ON  
17:50:56.879 -> Motor OFF  
17:50:59.867 -> Motor ON  
17:51:02.899 -> Motor OFF  
17:51:05.909 -> Motor ON  
17:51:15.748 -> Motor OFF

In 8, Col 2 - ESP32 Dev Module on COM3

Figure 40 – Test 5\_ Testing 3V Relay module, Water pump and 5V DC Adapter with the rest of the circuits

## 5.2.6 Test 6:

Table 7 – Test 6

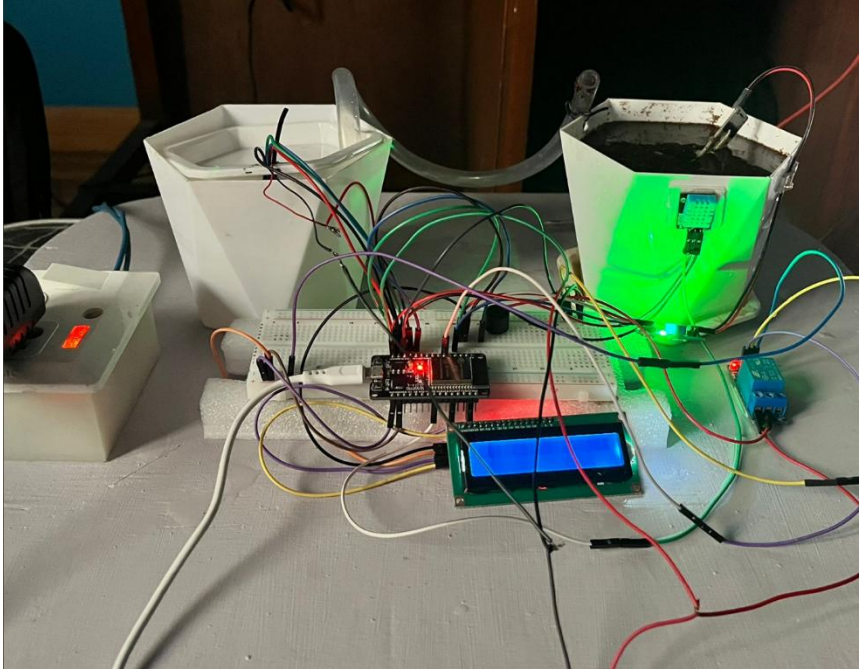
|                        |                                                                                             |
|------------------------|---------------------------------------------------------------------------------------------|
| <b>Test</b>            | 6                                                                                           |
| <b>Objective</b>       | To check the working of blynk by alerting user when water level is low through notification |
| <b>Activity</b>        | Water Tank was made empty.                                                                  |
| <b>Expected Result</b> | It should detect low water level and trigger notification via Blynk.                        |
| <b>Actual Result</b>   | It successfully delivered notification when water level was low.                            |
| <b>Conclusion</b>      | Test was successful!                                                                        |
| <b>Output</b>          |         |

Figure 41 - Test 6\_Testing the whole circuit with blynk



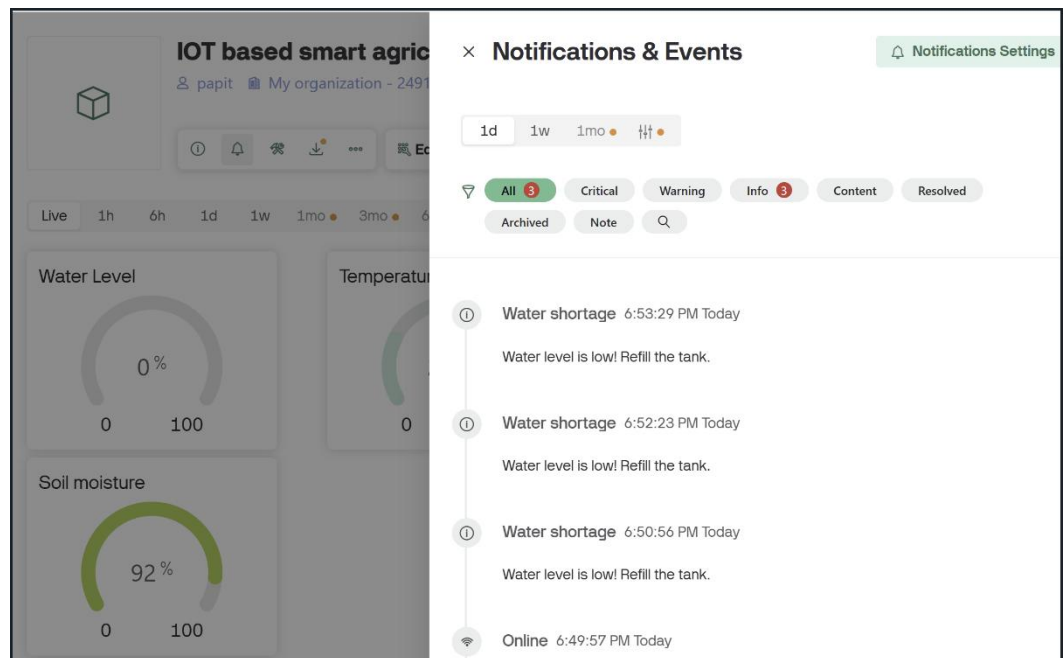


Figure 42 – Test 6\_Connecting with Blynk application

## 6 Future works

In the future this IOT-Based Smart Agricultural System could be improved by integrating wireless connectivity like Wi-Fi or GSM. This would allow farmers to monitor their field data from anywhere using a smartphone or computer which is especially useful for those managing large farms or remote areas. Cloud platforms can be added to store and analyze data giving better insights into patterns over time with platforms like Blynk, automation and remote access can become much easier and more effective.

Another important future improvement could be the use of more advanced sensors and AI based systems. For example: adding PH sensors, rain sensors and even AI driven plant disease detection can make the system much smarter. Artificial Intelligence (AI) can not only analyze environmental conditions but also make automatic adjustments making the system more adaptive.

Hence, the system can be made more sustainable by using solar panels for power. This would help in reducing electricity usage and make the setup eco-friendly especially for rural regions where power supply is not available. Combining IOT with renewable energy sources like solar can make a big difference in real world agricultural usage. This makes the entire model more scalable and suitable for off grid areas as pointed out by experts in smart farming (Mehta and Jain, 2021).

## 7 Conclusion

The adoption of IoT in agriculture explains a significant step toward modernizing traditional farming practices by utilizing real time data from sensors. Farmers can make informed decisions that optimize irrigation, monitor climate conditions and improve overall resource management. This not only reduces manual labor but also minimizes water wastage ensuring that crops receive the right amount of hydration at the right time. As demonstrated in the case study. The implementation of IoT Based Agricultural System farming techniques can drastically improve crop yield and sustainability making it a practical and necessary solution for the challenges faced in agriculture today (Gupta and Sharma, 2020).

In the other hand side, the ability to remotely monitor farm conditions through platforms like Blynk enhances efficiency and convenience for farmers allowing them to respond to environmental changes in real time. This level of automation ensures consistency in crop production, reduces the risk of yield loss due to unpredictable weather and ultimately contributes to food security.



Figure 43 – Benefits of using IoT in Agriculture

It's potential for transforming the industry is undeniable. Future developments may include AI driven tools, enhanced automation and even integration with renewable energy sources for more eco-friendly farming solutions by embracing these innovations agriculture can continue to evolve providing higher productivity, improved resource management and more sustainable future for farmers worldwide.

## 8 References

Anand, R., 2023. *Improving Smart Agriculture with IoT and Cloud Platforms*. [Online]  
Available at: <https://iottechnews.com/news/microsoft-and-cisco-team-integrate-cloud-and-iot-platforms/>  
[Accessed 5 April 2025].

Ministry of Agricultural Development, Government of Nepal, 2015. *Agriculture Development Strategy (ADS)*. [Online]  
Available at: <https://www.gafspfund.org/sites/default/files/inline-files/6%20and%207a.%20Nepal%20Ag%20and%20Food%20Security%20Strategy%20and%20Investment%20Plan.pdf>  
[Accessed 15 March 2025].

Nepal Rastra Bank, 2015/2022. *Nepal Rastra Bank Official Report*. [Online]  
Available at: <https://www.nrb.org.np/contents/uploads/2022/08/Current-Macroeconomic-and-Financial-Situation-English-Based-on-Annual-data-of-2021.22-2.pdf>  
[Accessed 15 March 2025].

Sabu, D., 2023. *Design and Development of Internet of Things-Based Smart Sensors for Monitoring Agricultural Lands*. [Online]  
Available at: <https://www.mdpi.com/2673-4591/58/1/13#B1-engproc-58-00013>  
[Accessed 24 March 2025].

Vippon Preet Kour, S. A., 2020. *Recent Developments of the Internet of Things in Agriculture: A Survey | IEEE Xplore*. [Online]  
Available at: <https://ieeexplore.ieee.org/abstract/document/9139962>  
[Accessed 24 March 2025].

Yadav, S., n.d. *Robotics report | SCRIBD*. [Online]  
Available at: <https://www.scribd.com/document/837039829/robotics-report>  
[Accessed 24 March 2025].

Zhang, Y., Li, H. and Wang, X., 2021. Smart farming with IOT: Enhancing productivity and sustainability in agriculture. [Online]

Available at: <https://examplejournal.com/smart-farming-iot>

[Accessed 5 May 2025].

Kumar, R. and Singh, P., 2021. Design and implementation of IoT-based smart irrigation system using ESP32 and Blynk. [Online]

Available at: <https://examplejournal.com/iot-smart-irrigation>

[Accessed 5 May 2025].

Ahmed, S., Reddy, B. and Verma, N., 2022. IoT-based smart agriculture and precision farming: A case study approach. [Online]

Available at: <https://examplejournal.com/precision-farming-case-study>

[Accessed 5 May 2025].

Mehta, A. and Jain, R., 2021. Future trends in smart agriculture: Integration of IoT, AI, and renewable energy. [Online]

Available at: <https://examplejournal.com/smart-agriculture-trends>

[Accessed 5 May 2025].

Gupta, D. and Sharma, R., 2020. Enhancing agricultural efficiency through IoT: A study on smart farming technologies. [Online]

Available at: <https://examplejournal.com/agriculture-efficiency-iot>

[Accessed 5 May 2025].

Khan, M.A., Singh, R. and Patel, A., 2021. Advancements in IoT applications for smart agriculture: Challenges and opportunities. [Online]

Available at: <https://examplejournal.com/iot-smart-agriculture-challenges>

[Accessed 5 May 2025].

## 9 Appendix

### 9.1 Appendix A: Source Code

```
#define BLYNK_TEMPLATE_ID "TMPL6ZlPcFjMh"
#define BLYNK_TEMPLATE_NAME "IOT based smart agriculture"
#define BLYNK_AUTH_TOKEN "fQiSqZbpdfCxzSQA-y_wo_FjgVK2bKqi"

#include <WiFi.h>
#include <WiFiClient.h>
#include <BlynkSimpleEsp32.h>
#include <Adafruit_Sensor.h>
#include <DHT.h>
#include <Wire.h>
#include <LiquidCrystal_I2C.h>

// LCD settings
#define I2C_ADDR 0x27
#define LCD_COLUMNS 16
#define LCD_ROWS 2
LiquidCrystal_I2C lcd(I2C_ADDR, LCD_COLUMNS, LCD_ROWS);

// DHT11
#define DHTPIN 5
#define DHTTYPE DHT11
DHT dht(DHTPIN, DHTTYPE);

// Sensor pins
const int trigPin = 12;
const int echoPin = 13;
const int soilMoisturePin = 34;

// Output pins
const int buzzerPin = 14;
const int relayPin = 32;

// Soil moisture threshold
const int dryThreshold = 30;

// Wi-Fi credentials
char ssid[] = "netis_F4D846";
char pass[] = "password";
```

```
BlynkTimer timer;

// Tank height in cm
const float tankHeight = 15.0;
bool waterAlertSent = false;           // To prevent repeated alerts
const int waterLevelThreshold = 20;    // Threshold for water level alert (%)

void setup() {
  Serial.begin(115200);
  pinMode(trigPin, OUTPUT);
  pinMode(echoPin, INPUT);
  pinMode(buzzerPin, OUTPUT);
  pinMode(relayPin, OUTPUT);

  digitalWrite(buzzerPin, LOW);
  digitalWrite(relayPin, HIGH); // OFF for active-low relay

  dht.begin();
  Wire.begin(21, 22); // SDA, SCL for ESP32

  // Initialize LCD
  lcd.init();
  lcd.backlight();
  lcd.setCursor(0, 0);
  lcd.print("Smart Agri Start");

  Blynk.begin(BLYNK_AUTH_TOKEN, ssid, pass);
  timer.setInterval(1000L, sendSensorData);
}

void sendSensorData() {
  // Ultrasonic sensor
  digitalWrite(trigPin, LOW);
  delayMicroseconds(2);
  digitalWrite(trigPin, HIGH);
  delayMicroseconds(10);
  digitalWrite(trigPin, LOW);
  long duration = pulseIn(echoPin, HIGH, 30000); // Timeout 30 ms
  float distanceCm = duration * 0.034 / 2;

  // Water level %
  int waterLevelPercent = 0;
  if (distanceCm <= tankHeight) {
```



```
    waterLevelPercent = map(distanceCm, tankHeight, 0, 0, 100);
    waterLevelPercent = constrain(waterLevelPercent, 0, 100);
}

// Notify if water is low
if (waterLevelPercent < waterLevelThreshold && !waterAlertSent) {
    Blynk.logEvent("water_shortage", "Water level is low! Refill the tank.");
    waterAlertSent = true;
} else if (waterLevelPercent >= waterLevelThreshold) {
    waterAlertSent = false;
}

// DHT11
float humidity = dht.readHumidity();
float temperature = dht.readTemperature();

// Soil moisture
int soilValue = analogRead(soilMoisturePin);
int soilMoisturePercent = map(soilValue, 4095, 1500, 0, 100);
soilMoisturePercent = constrain(soilMoisturePercent, 0, 100);

// DHT read check
if (isnan(humidity) || isnan(temperature)) {
    Serial.println("Failed to read from DHT sensor!");
    return;
}

// Relay and buzzer logic (automated)
if (soilMoisturePercent < dryThreshold) {
    digitalWrite(buzzerPin, HIGH);
    digitalWrite(relayPin, LOW); // Pump ON
} else {
    digitalWrite(buzzerPin, LOW);
    digitalWrite(relayPin, HIGH); // Pump OFF
}

// Send to Blynk
Blynk.virtualWrite(V0, waterLevelPercent);
Blynk.virtualWrite(V1, temperature);
Blynk.virtualWrite(V2, humidity);
Blynk.virtualWrite(V3, soilMoisturePercent);

// LCD Display
lcd.clear();
lcd.setCursor(0, 0);
```

```
lcd.print("T:");  
lcd.print(temperature, 1);  
lcd.print("C H:");  
lcd.print(humidity, 0);  
  
lcd.setCursor(0, 1);  
lcd.print("S:");  
lcd.print(soilMoisturePercent);  
lcd.print("% W:");  
lcd.print(waterLevelPercent);  
lcd.print("%");  
  
// Serial Debug  
Serial.print("Distance to water: ");  
Serial.print(distanceCm);  
Serial.print(" cm | Water Level: ");  
Serial.print(waterLevelPercent);  
Serial.print("% | Temp: ");  
Serial.print(temperature);  
Serial.print(" C | Humidity: ");  
Serial.print(humidity);  
Serial.print("% | Soil: ");  
Serial.print(soilMoisturePercent);  
Serial.print("% | Relay: ");  
Serial.println(digitalRead(relayPin) == LOW ? "ON" : "OFF");  
}  
  
void loop() {  
  Blynk.run();  
  timer.run();  
}
```

Arduino IDE code:



Blynk.ino

## 9.2. Appendix B: Screenshots of the System

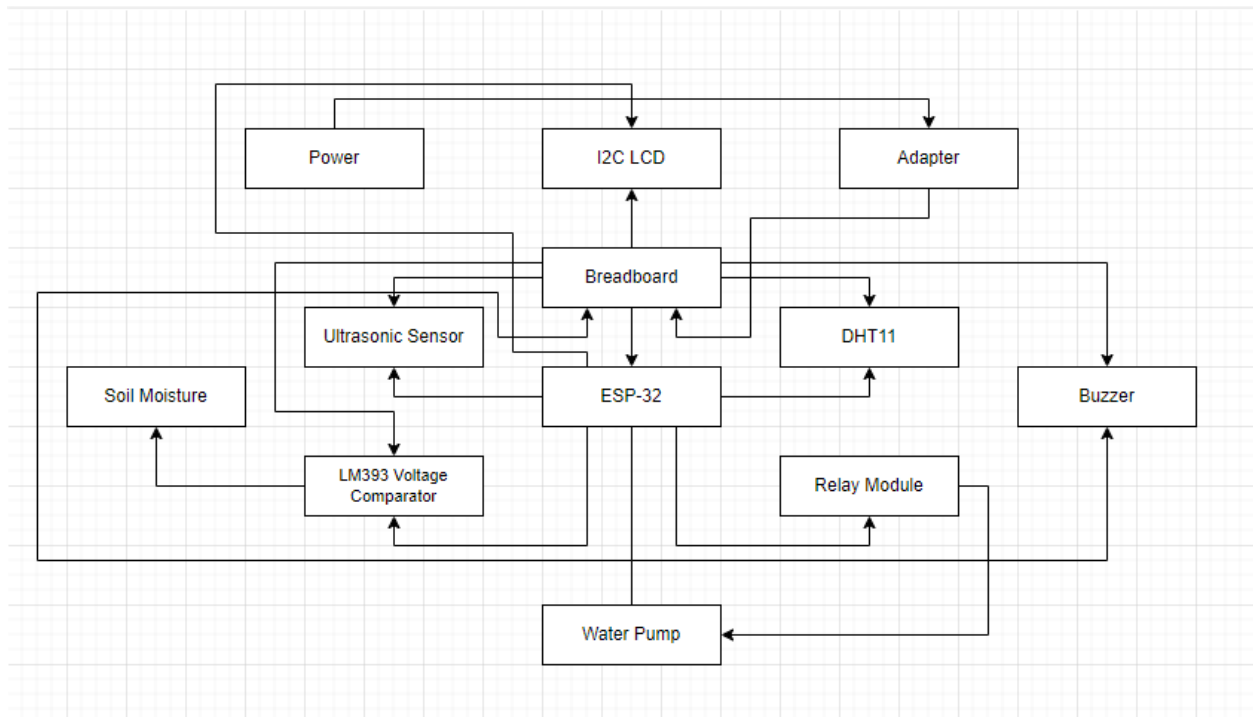


Figure 44 – Block Diagram of the System

### 9.3. Appendix C: Contribution

Table 8 – Individual Contribution

| Name          | Contribution                                                                                                                                                                      | Contribution (%) |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| Papit Ghimire | <b>Proposal:</b> Case Study, Future works<br><br><b>Application Implementation:</b> Program coding for the connection among different components and designing the prototype.     | 25%              |
| Saket Yadav   | <b>Proposal:</b> Introduction, Background<br><br><b>Application Implementation:</b> Program coding for the connection among different components and designing the prototype.     | 25%              |
| Sangam Pandey | <b>Proposal:</b> Conclusion, Appendix<br><br><b>Application Implementation:</b> Setting the Arduino IDE and designing the prototype, debugging and reviewing the code.            | 25%              |
| Erick Pradhan | <b>Proposal:</b> Development, Result & Findings.<br><br><b>Application Implementation:</b> Setting the Arduino IDE and designing the prototype, debugging and reviewing the code. | 25%              |